COHERENT QUADRANT

Global Medical Films Market

Global

150 Manufacturer scored on Product Capability (X) and Business Capability (Y)

Market Classification

Market

Global Medical Films Market

Market Type

This landscape covers the suppliers of the medical-grade POLYMER MATERIALS that medical films are made from: thermoplastic polyurethane (TPU) and other polyurethanes, polyethylene, polypropylene, polyester / PET, ethylene vinyl acetate, silicone-based polymers, polyamide and other medical-grade film resins, compounds and elastomers, sold to medical film converters, adhesive and lamination makers and device makers for wound dressing, catheter securement, surgical drape, NPWT, wearable sensor and medical packaging films.

Provider Categories in this Market

1. Manufacturer

Manufacturer is the chosen role because the competitive landscape is defined by companies producing and commercializing their own physical, medical-grade film materials rather than providing software or pure services.

150	20	20	4
Companies assessed	Companies on the Coherent Quadrant	Scoring parameters	Quadrants

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Market Scoring Parameters

X — Product Capability

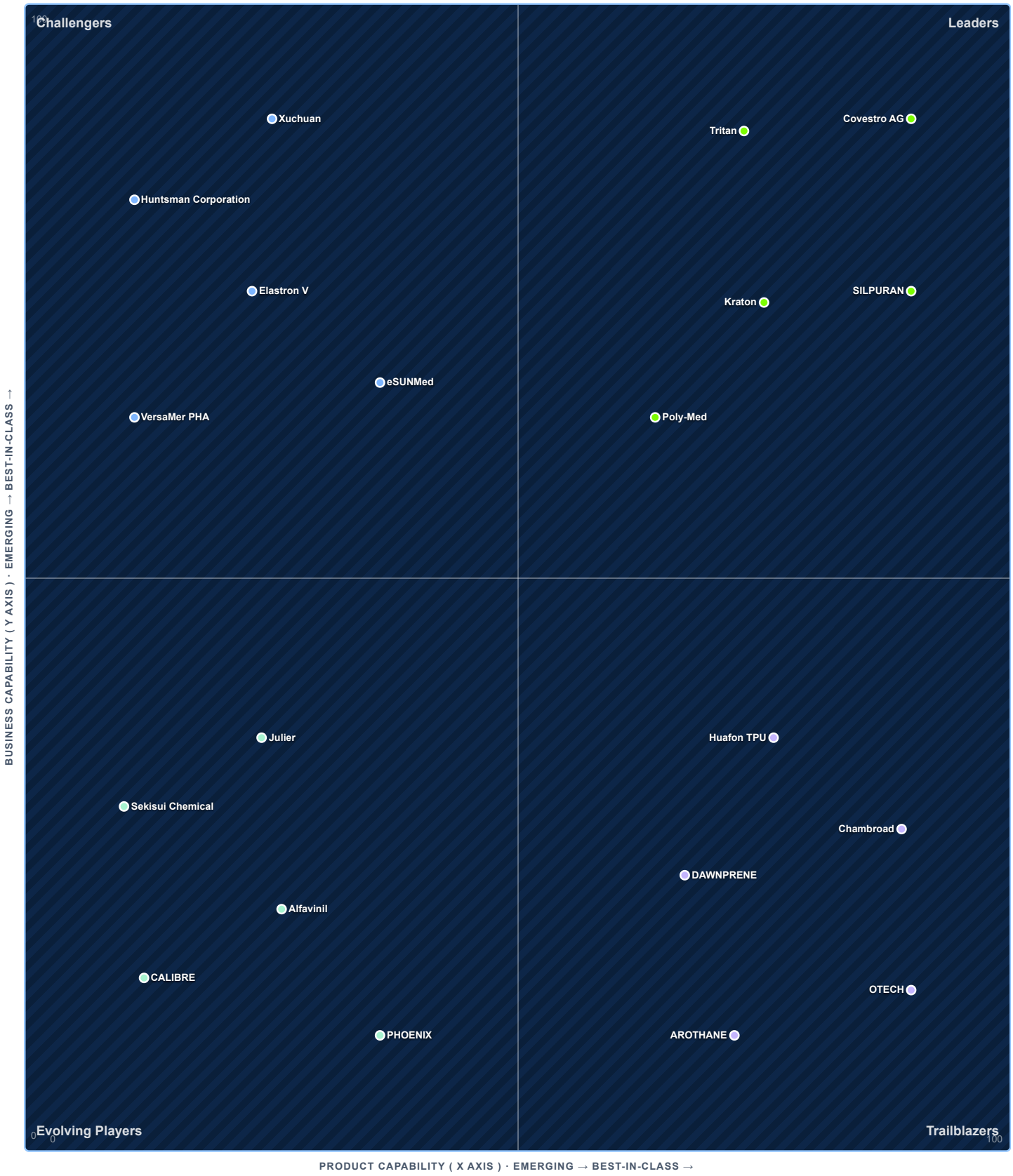
1. **Medical-Grade Polymer Portfolio Breadth**
2. **Film Performance & Functional Capabilities**
3. **Regulatory Compliance & Biocompatibility**
4. **Customization & Co-Development Capability**
5. **Manufacturing Quality & Consistency**

Y — Business Capability

1. **Medical Film Revenue & Growth**
2. **Named Customers & Contract Wins**
3. **Global Production & Supply Footprint**
4. **R&D Investment & Innovation Pipeline**
5. **Market Share & Brand Leadership**

Quadrant Positioning

(20 on chart · 150 in total)



Countries on chart (by quadrant)

Leaders: Germany×2, United States×3 · **Challengers**: China×2, Turkey×1, United States×1, Canada×1 · **Trailblazers**: China×3, United States×1, South Korea×1 · **Evolving Players**: Taiwan×1, Argentina×1, China×1, United States×1, Japan×1

What Each Quadrant Means

Leaders **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

Challengers **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

Trailblazers **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

Emerging Players **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

Top 20 Companies

(plotted on the Quadrant Positioning chart)

BRAND	COMPANY	HQ	ROLE	QUADRANT	X (PRODUCT CAPABILITY)	Y (BUSINESS CAPABILITY)	OVERALL SCORE
Covestro AG	Covestro AG (acquired by XRG (ADNOC investment subsidiary))	Leverkusen, Germany	Manufacturer	Leaders	96	96	96
Tritan	Eastman Chemical Company	Kingsport, Tennessee, USA	Manufacturer	Leaders	92	93	92
SILPURAN	Wacker Chemie AG	Munich, Germany	Manufacturer	Leaders	92	92	92
Kraton	Kraton Corporation (subsidiary of DL Chemical)	Houston, Texas, USA	Manufacturer	Leaders	90	84	87
Poly-Med	Poly-Med, Inc.	Anderson, South Carolina, USA	Manufacturer	Leaders	87	81	84
Xuchuan	Xuchuan Chemical (Suzhou) Co., Ltd.	Suzhou, China	Manufacturer	Challengers	80	85	82
Elastron V	Elastron Kimya San. ve Tic. A.Ş.	Kocaeli, Turkey	Manufacturer	Challengers	82	79	80
Huntsman Corporation	Huntsman Corporation	The Woodlands, United States	Manufacturer	Challengers	79	80	80
eSUNMed	eSUNMed Biotechnology (Shenzhen) Co., Ltd.	Shenzhen, China	Manufacturer	Challengers	82	76	79
VersaMer PHA	TerraVerdae Bioworks Inc.	Edmonton, Canada	Manufacturer	Challengers	76	76	76
Chambroad	Shandong Chambroad Holding Group Co., Ltd.	Binzhou, China	Manufacturer	Trailblazers	85	79	82
Huafon TPU	Zhejiang Huafon Thermoplastic Polyurethane Co., Ltd.	Rui'an, China	Manufacturer	Trailblazers	84	81	82
OTECH	OTECH Corporation	Rolling Prairie, Indiana, USA	Manufacturer	Trailblazers	86	74	80
DAWNPRENE	Shandong Dawn Polymer Co., Ltd.	Longkou, China	Manufacturer	Trailblazers	83	74	78
AROTHANE	CK Medience	Seoul, South Korea	Manufacturer	Trailblazers	84	70	77
PHOENIX	Phon Tech Industrial Company	Douliu City, Taiwan	Manufacturer	Evolving Players	83	72	78
Alfavinil	Alfavinil S.A.	Buenos Aires, Argentina	Manufacturer	Evolving Players	82	73	78
Julier	Julier (Xiamen) Technology Co., Ltd.	Xiamen, China	Manufacturer	Evolving Players	81	74	78
CALIBRE	Trinseo PLC	Wayne, Pennsylvania, USA	Manufacturer	Evolving Players	79	72	76
Sekisui Chemical	SEKISUI CHEMICAL CO., LTD.	Osaka, Japan	Manufacturer	Evolving Players	77	73	75

Other Noticeable Players

(130 companies)

BRAND	COMPANY	HQ	ROLE	RELATIVE POSITIONING	STRENGTH
GUR	Celanese Corporation	Irving, Texas, USA	Manufacturer	Leaders	
DOW HEALTH+	Dow Inc.	Midland, Michigan, USA	Manufacturer	Leaders	
THERMOLAST M	KRAIBURG TPE GmbH & Co. KG	Waldkraiburg, Germany	Manufacturer	Leaders	
SKYGREEN	SK chemicals	Seongnam, South Korea	Manufacturer	Leaders	
Panlite	Teijin Limited	Tokyo, Japan	Manufacturer	Leaders	
Techmer PM	Techmer PM (subsidiary of Gryphon Investors)	Clinton, Tennessee, USA	Manufacturer	Leaders	
Toray	Toray Industries, Inc.	Tokyo, Japan	Manufacturer	Leaders	
Defend Rx	Compounding Solutions LLC	Lewiston, Maine, USA	Manufacturer	Leaders	
BASF SE	BASF SE	Ludwigshafen, Germany	Manufacturer	Leaders	
KRAIBURG TPE	KRAIBURG TPE	Waldkraiburg, Germany	Manufacturer	Leaders	
Kuraray Co., Ltd.	Kuraray Co., Ltd.	Tokyo, Japan	Manufacturer	Leaders	
Silopren	Momentive Performance Materials Inc. (acquired by KCC Corporation)	Niskayuna, New York, USA	Manufacturer	Leaders	
GreenMed	Nan Ya Plastics Corp.	Kaohsiung, Taiwan	Manufacturer	Leaders	
Wanhua Chemical Group Co., Ltd.	Wanhua Chemical Group Co., Ltd.	Yantai, China	Manufacturer	Leaders	
Americhem	Americhem, Inc. (acquired by Pritzker Private Capital)	Cuyahoga Falls, Ohio, USA	Manufacturer	Leaders	
MedEco	BIOVOX GmbH	Darmstadt, Germany	Manufacturer	Leaders	
APEL	Mitsui Chemicals, Inc.	Tokyo, Japan	Manufacturer	Leaders	
meliflex XP	MELITEK A/S	Nr. Alslev, Denmark	Manufacturer	Leaders	
Clyrell	HMC Polymers Co., Ltd.	Bangkok, Thailand	Manufacturer	Leaders	
DIAMIRON	Mitsubishi Chemical Corporation	Tokyo, Japan	Manufacturer	Leaders	
The Lubrizol Corporation	The Lubrizol Corporation (subsidiary of Berkshire Hathaway Inc.)	Wickliffe, United States	Manufacturer	Leaders	
ICP DAS CO., LTD.	ICP DAS CO., LTD.	Hsinchu County, Taiwan	Manufacturer	Leaders	
TERLUX HD	INEOS Styrolution Group GmbH	Frankfurt, Germany	Manufacturer	Leaders	
Extrulen	Mitsubishi Chemical Advanced Materials GmbH (subsidiary of Mitsubishi Plastics)	Vreden, Germany	Manufacturer	Leaders	
Riken Technos	Riken Technos Corporation	Tokyo, Japan	Manufacturer	Leaders	
EVERLON	Ever Polymer Co., Ltd.	Taoyuan City, Taiwan	Manufacturer	Leaders	
SCGC	SCG Chemicals Public Company Limited	Bangkok, Thailand	Manufacturer	Leaders	
ECOSYAR	Toyobo Co., Ltd.	Osaka, Japan	Manufacturer	Leaders	
Array	Alpek, S.A.B. de C.V.	San Pedro Garza Garcia, Mexico	Manufacturer	Leaders	
COIM Group	COIM Group	Offanengo, Italy	Manufacturer	Leaders	
Sinopec	China Petroleum & Chemical Corporation	Beijing, China	Manufacturer	Leaders	
Topilene	Hyosung Chemical	Seoul, South Korea	Manufacturer	Leaders	
Teknor Apex Company	Teknor Apex Company	Pawtucket, United States	Manufacturer	Leaders	

Avient Corporation	Avient Corporation	Avon Lake, United States	Manufacturer	Leaders	
P-TEX	Formosa Plastics Corporation	Kaohsiung, Taiwan	Manufacturer	Leaders	
ISOTHANE	Great Eastern Resins Industrial Co., Ltd.	Taichung, Taiwan	Manufacturer	Leaders	
EXCELLEN	Sumitomo Chemical Co., Ltd.	Tokyo, Japan	Manufacturer	Leaders	
Mirathane	Miracll Chemicals Co., Ltd.	Yantai, China	Manufacturer	Leaders	
ALPHAMED	Alphagary Compuestos	Mexico City, Mexico	Manufacturer	Leaders	
LUPURE	LG Chem	Seoul, South Korea	Manufacturer	Leaders	
starex	LOTTE Chemical	Seoul, South Korea	Manufacturer	Leaders	
Arkema	Arkema	Puteaux, France	Manufacturer	Leaders	
Hanwha TotalEnergies	Hanwha TotalEnergies Petrochemical Co., Ltd.	Daesan, South Korea	Manufacturer	Leaders	
ViviOn	USI Corporation	Taipei, Taiwan	Manufacturer	Leaders	
EMBLEM	Unitika Ltd.	Osaka, Japan	Manufacturer	Leaders	
RTP Company	RTP Company	Winona, United States	Manufacturer	Leaders	
Foster	Foster Corporation (acquired by GEON Performance Solutions)	Putnam, Connecticut, USA	Manufacturer	Leaders	
Medcol	Braskem S.A.	Sao Paulo, Brazil	Manufacturer	Leaders	
VECTOR	TSRC Corporation	Taipei, Taiwan	Manufacturer	Leaders	
Globalene	LCY Chemical Corp. (subsidiary of KKR)	Taipei, Taiwan	Manufacturer	Leaders	
HEXPOL TPE	HEXPOL TPE	Malmö, Sweden	Manufacturer	Leaders	
Hi-Dura	Ascend Performance Materials LLC (subsidiary of SK Capital Partners)	Houston, Texas, USA	Manufacturer	Challengers	
TOPAS	TOPAS Advanced Polymers GmbH (subsidiary of Polyplastics Co., Ltd.)	Frankfurt am Main, Germany	Manufacturer	Leaders	
Zeonor	Zeon Corporation	Tokyo, Japan	Manufacturer	Leaders	
TEAMLONG	Changzhou Teamlong Medical Polymer Co., Ltd.	Changzhou, China	Manufacturer	Trailblazers	
Marlex	Chevron Phillips Chemical Company LLC	The Woodlands, Texas, USA	Manufacturer	Leaders	
KLJ	KLJ Polymers & Chemicals Limited	New Delhi, India	Manufacturer	Leaders	
VYTHENE	Alphagary (subsidiary of Orbia)	Boston, United States	Manufacturer	Leaders	
Formolene	Formosa Plastics Corporation, U.S.A.	Livingston, New Jersey, USA	Manufacturer	Trailblazers	
Shuang-Bang	Shuang-Bang Industrial Corp.	Nantou City, Taiwan	Manufacturer	Challengers	
XJY Silicones	Jiangxi New Jiayi New Materials Co., Ltd.	Jiujiang, China	Manufacturer	Challengers	
Linghua	Yantai Linghua New Material Co., Ltd.	Yantai, China	Manufacturer	Challengers	
Genvan	Shenzhen Genvan Silicone Materials Co., Ltd.	Shenzhen, China	Manufacturer	Challengers	
TAIRIREX	Formosa Chemicals & Fibre Corp.	Taipei, Taiwan	Manufacturer	Evolving Players	
Alcom MED	MOCOM Compounds GmbH & Co. KG	Hamburg, Germany	Manufacturer	Challengers	
Shin-Etsu	Shin-Etsu Chemical Co., Ltd.	Tokyo, Japan	Manufacturer	Trailblazers	
PVCL	PVC Colouring Compounding & Processing	Ahmedabad, India	Manufacturer	Challengers	
Coating P. Materials	Coating P. Materials Co., Ltd.	Taichung, Taiwan	Manufacturer	Challengers	
DIC	DIC Corporation	Tokyo, Japan	Manufacturer	Challengers	
Miractran	Japan Miractran Co., Ltd.	Tokyo, Japan	Manufacturer	Evolving Players	

HRS	HRS Co., Ltd.	Seoul, South Korea	Manufacturer	Evolving Players	
New Shanghua	Shanghai New Shanghua Polymer Material Co., Ltd.	Shanghai, China	Manufacturer	Challengers	
AM Masterbatch	AM Masterbatch Pvt. Ltd.	Ahmedabad, India	Manufacturer	Challengers	
BARRIALON	Asahi Kasei Corporation	Tokyo, Japan	Manufacturer	Trailblazers	
Newsil	Dongguan Newsil Silicone Technology Co., Ltd.	Dongguan, China	Manufacturer	Evolving Players	
Hanwha	Hanwha Solutions Chemical Division	Seoul, South Korea	Manufacturer	Trailblazers	
Titanpro	Lotte Chemical Titan (M) Sdn Bhd	Kuala Lumpur, Malaysia	Manufacturer	Evolving Players	
Savvy	Savvy Enterprises	Faridabad, India	Manufacturer	Challengers	
InnoPlus	PTT Global Chemical Public Company Limited	Bangkok, Thailand	Manufacturer	Evolving Players	
Audia Elastomers	Audia Elastomers	LaFayette, Georgia, USA	Manufacturer	Evolving Players	
KEBABLEND	BARLOG Plastics GmbH	Overath, Germany	Manufacturer	Trailblazers	
Kin Join	Kin Join Co., Ltd.	Taoyuan City, Taiwan	Manufacturer	Evolving Players	
Littleflower	Jiangsu Kaier High Polymer Materials Co., Ltd.	Yangzhou, China	Manufacturer	Evolving Players	
KOSLEN	Kunsun Polymer Material (Fujian) Co., Ltd.	Quanzhou, China	Manufacturer	Evolving Players	
SiSiB	Nanjing SiSiB Silicones Co., Ltd.	Nanjing, China	Manufacturer	Evolving Players	
Silibase Silicone	Silibase Silicone	Jiande, China	Manufacturer	Evolving Players	
LUVOCOM	Lehmann & Voss & Co. KG	Hamburg, Germany	Manufacturer	Evolving Players	
CURESORB	CURE BIOTECH Co., Ltd.	Cheongju-si, South Korea	Manufacturer	Challengers	
FUTHANE	Taiwan Free Radical Performance Materials Co., Ltd.	Nantou, Taiwan	Manufacturer	Evolving Players	
Aronvis	Toagosei Co., Ltd.	Tokyo, Japan	Manufacturer	Evolving Players	
Orion	Star Plastics, Inc. (subsidiary of Akoya Capital Partners)	Ravenswood, West Virginia, USA	Manufacturer	Evolving Players	
Halene-P	Haldia Petrochemicals Limited	Kolkata, India	Manufacturer	Trailblazers	
KK Kompounding	KK Kompounding Tech Giant Ltd	Mumbai, India	Manufacturer	Evolving Players	
Iupilon	Mitsubishi Engineering-Plastics Corporation	Tokyo, Japan	Manufacturer	Trailblazers	
Capa	Ingevity Corporation	North Charleston, South Carolina, USA	Manufacturer	Evolving Players	
BabanPlastic	BabanPlastic	Aurangabad, India	Manufacturer	Evolving Players	
POLIMAXX	IRPC Public Company Limited	Bangkok, Thailand	Manufacturer	Evolving Players	
Vinyltec Industry	Vinyltec Industry Co., Ltd.	Samut Prakan, Thailand	Manufacturer	Evolving Players	
Pelprene	Toyobo MC Corporation	Osaka, Japan	Manufacturer	Evolving Players	
Symflex S	Symplast Elastomer Tasarım San. ve Tic. A.Ş.	Izmir, Turkey	Manufacturer	Evolving Players	
Humstar	Dongguan Humstar Macromolecule Technology Co., Ltd.	Dongguan, China	Manufacturer	Evolving Players	
Perfect Colourants & Plastics	Perfect Colourants & Plastics Pvt. Ltd.	Vadodara, India	Manufacturer	Evolving Players	
SILIKE	Chengdu Silike Technology Co., Ltd.	Chengdu, China	Manufacturer	Evolving Players	
ZINCA	Anhui Zinca Silicone Technologies Co., Ltd.	Bengbu, China	Manufacturer	Evolving Players	
KCC SILICONE	KCC Silicone Corporation	Seoul, South Korea	Manufacturer	Evolving Players	
Ocean Plastics	Ocean Plastics Co., Ltd.	Taipei, Taiwan	Manufacturer	Evolving Players	
Renk Granül	Renk Granül San. ve Tic. A.Ş.	Istanbul, Turkey	Manufacturer	Evolving Players	

Betely	Shenzhen Betely Polymer Materials Co., Ltd.	Shenzhen, China	Manufacturer	Evolving Players	
KOSYN	Kumho Petrochemical Co., Ltd.	Seoul, South Korea	Manufacturer	Evolving Players	
MOSTEN	ORLEN Unipetrol RPA, s.r.o.	Prague, Czechia	Manufacturer	Evolving Players	
EKP-MG55E	Ekmen Plastik PVC Granül San. ve Tic. Ltd. Şti.	Istanbul, Turkey	Manufacturer	Evolving Players	
SILON	SILON s.r.o.	Prague, Czech Republic	Manufacturer	Evolving Players	
ProvaMed	ACTEGA GmbH	Bremen, Germany	Manufacturer	Evolving Players	
CHANGYOUNG Chemical	CHANGYOUNG Chemical Co., Ltd.	Eumseong-gun, South Korea	Manufacturer	Evolving Players	
Flexon Polybend	Flexon Polybend	Rajkot, India	Manufacturer	Evolving Players	
Gemapilen	Gema Polimer Plastik Ürünleri San. ve Tic. A.Ş.	Istanbul, Turkey	Manufacturer	Evolving Players	
Advanced Polymer Materials	Advanced Polymer Materials Inc.	Montreal, Canada	Manufacturer	Evolving Players	
ADB	Applied DB Public Company Limited	Samut Prakan, Thailand	Manufacturer	Evolving Players	
Zhengtai	Dongguan Zhengtai Organic Silicon Technology Co., Ltd.	Dongguan, China	Manufacturer	Evolving Players	
Epaflex Sheet	Epaflex Polyurethanes S.r.l. (subsidiary of Elachem S.p.A.)	Cassolnovo, Italy	Manufacturer	Evolving Players	
Quality Polymers	Quality Polymers	Vasai, India	Manufacturer	Evolving Players	
Sunko Ink	Sunko Ink Co., Ltd.	Taichung, Taiwan	Manufacturer	Evolving Players	
TECATEC	Ensinger GmbH	Nufringen, Germany	Manufacturer	Trailblazers	
Green PVC	Green PVC Joint Stock Company	Phu Tho Province, Vietnam	Manufacturer	Evolving Players	
Hopefinder	Shenzhen Career Industrial Co., Ltd.	Shenzhen, China	Manufacturer	Evolving Players	
Jieh-Ming	Jieh-Ming Plastics Mfg. Co., Ltd.	New Taipei City, Taiwan	Manufacturer	Evolving Players	
Genecis Bioindustries	Genecis Bioindustries Inc.	Toronto, Canada	Manufacturer	Evolving Players	
Huadai	Dongguan Huadai Silicone Co., Ltd.	Dongguan, China	Manufacturer	Evolving Players	
AuroraGuard	Aurora Material Solutions, LLC (subsidiary of Nautic Partners)	Streetsboro, Ohio, USA	Manufacturer	Evolving Players	
SONGSTOMER	SONGWON Industrial Co., Ltd.	Ulsan, South Korea	Manufacturer	Evolving Players	

Parameter Detail by Company

(Product Capability and Business Capability evidence for the 20 companies on the chart)

Covestro AG (acquired by XRG (ADNOC investment subsidiary)) Covestro AG ● Leaders X 96 Y 96 Overall 96

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 93/100

> The rating rests on Covestro's established commercialized medical-grade material breadth spanning thermoplastic polyurethanes (TPU) and polycarbonate (PC) based configurations specifically formulated for medical films and device overlays. Based on primary research, the company's medical film-grade polymer offerings can be estimated as limited to a few core classes, based on its primary commercial focus on thermoplastic polyurethane and polycarbonate medical films.

SUPPORTING POINTS

- The company reports its dedicated medical portfolio includes Platilon and Dureflex thermoplastic polyurethane (TPU) films engineered for wound care and medical wearables, Bayfol TPU/PETG bi-layer films for medical packaging, and Makrofol polycarbonate (PC) films used in medical sterilization packaging, as highlighted at the MD&M West 2026 conference.
- Covestro launched a specific high-performance medical-grade TPU film product named Dureflex MA5000 in January 2026, which is explicitly commercialized for applications such as continuous glucose monitor (CGM) overpatches, IV securement dressings, and smart patches.

Film Performance & Functional Capabilities — scored 100/100

> We verified that the company has extensive published technical specifications demonstrating functional performance benchmarks for its Platilon and Dureflex medical-grade TPU product lines, including precise moisture vapor transmission rates and high-temperature thermal resistance. Based on primary research, the company's long-term performance degradation data can be estimated as extensive across multi-layer wear applications, based on its development of specialized films with high mechanical robustness, breathability, and skin-friendly properties for wound dressings and wearables.

SUPPORTING POINTS

- Covestro reports that its combination of hydrophilic Baymedix A and Platilon film yields a Moisture Vapor Transmission Rate (MVTR) of approximately 3,200 g/(day · m²) for a 55 µm adhesive layer under DIN EN 13726-2 Part 3.2 testing.
- Covestro announced Dureflex MA5000 in January 2026, a thermoplastic polyurethane (TPU) film engineered on a paper carrier that withstands high-temperature adhesive coatings above 135 °C while maintaining dimensional stability against distortion and wrinkling.

Regulatory Compliance & Biocompatibility — scored 100/100

> The rating rests on confirmed compliance statements showing that Covestro's primary medical film and polymer portfolios meet international biocompatibility criteria for short-term and skin-contact medical applications. Based on primary research, the company's laboratory verification certificates can be estimated as strong and fully compliant with major standard testing protocols, based on its public testing documentation confirming adherence to ISO 117 biocompatibility standards and USP Class VI compliance.

SUPPORTING POINTS

- The company reports in its medical reference documentation that its medical-grade thermoplastic polyurethane (TPU) portfolio meets ISO 10993 and USP Class VI standards for human tissue contact durations of 30 days or less.
- Covestro reports that its Bayfol MA multilayer films, designed specifically for orthopedic implant and medical packaging applications, are certified as compliant with both ISO 10993-5 and USP Class VI standards.

Customization & Co-Development Capability — scored 100/100

> We verified that Covestro actively engages in joint medical application design and provides customizable film specifications, including adjustable release forces and thicknesses, backed by its dedicated center of excellence for TPU films in Bomlitz, Germany. Based on primary research, the company's customer-turnaround times or drug-compatibility matrices can be estimated as robust for common clinical sterilization and chemical exposure methods, based on its published compatibility testing for common medical sterilization methods, antiseptics, and standard clinical fluids.

SUPPORTING POINTS

- Covestro collaborated with SunRay Scientific Inc. to co-develop a functional pulse oximeter demonstrator for medical wearables, combining Platilon TPU films with flexible printed electronics and lead-free conductive inks.
- The company engineered its customizable Platilon TPU film line with polypropylene carrier technology, allowing converters to tailor film thickness, surface finishes, colors, and liners according to specific customer parameters.

Manufacturing Quality & Consistency — scored 88/100

> The rating rests on Covestro's expansion of localized GMP-aligned medical TPU facilities and its adherence to ISO 9001 quality management systems across its healthcare material portfolios. Based on primary research, the company's polymer manufacturing lines certification or defect rates can be estimated as reliable and aligned with international medical manufacturing standards, based on its production footprint featuring specialized medical-grade facilities audited for stringent healthcare-sector quality management.

SUPPORTING POINTS

- The company reports that its medical-grade polycarbonates and thermoplastic polyurethane (TPU) portfolios are manufactured at ISO 9001 certified production sites that adhere to Good Manufacturing Practices (GMP) standards.
- The company expanded its localized production of medical-grade Desmopan Rx TPU to its Changhua facility in Taiwan, qualifying it as Covestro's second global plant—following the New Martinsville site in North America—equipped to manufacture these high-quality medical materials under explicit GMP principles.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 96/100

> We verified that the company's financial reporting consolidates its business into broad segments like Performance Materials and Solutions & Specialties without breaking out individual health sector totals. Based on primary research, the company's medical film revenue and growth can be estimated as growing steadily, based on its multi-million euro investments in expanding specialty film capacities globally and the strategic acquisition of Pontacol to extend its film business into medical technology.

SUPPORTING POINTS

- Covestro completed the acquisition of Pontacol AG in August 2025 to expand its specialty film portfolio into highly specialized multilayer adhesive films targeted toward medical technology, mobility, and textile industries.
- Covestro invested a low double-digit million-euro amount in October 2023 to expand its production capacity for the Platilon range of TPU films at its Bomlitz facility in Germany to support demand across applications including wound care.

Named Customers & Contract Wins — scored 93/100

> The rating rests on confirmed technical partnerships and intermediate material supply collaborations for medical device components. Based on primary research, the company's direct commercial supply contracts can be estimated as limited to a few markets, based on its publicly named smart patch innovation partnership with accensors and specialized medical material portfolio including Baymedix and Platilon brands.

SUPPORTING POINTS

- Covestro officially announced a technical joint partnership with SunRay Scientific Inc. to develop a fully functional pulse oximeter demonstrator utilizing its Platilon thermoplastic polyurethane (TPU) medical films alongside lead-free conductive inks.
- Covestro reports a commercialized materials supply collaboration with Mayser GmbH & Co. KG, which utilizes Covestro's Baymedix FD103 and AD111 aqueous polyurethane raw materials to manufacture its biocompatible, thermoformable "Mayser BLUEFOAM DG-T" used by original equipment manufacturers (OEMs) in the medical device market.

Global Production & Supply Footprint — scored 100/100

> We verified that the company possesses a concrete and expanding multi-regional footprint with production sites in Europe, Asia-Pacific, and North America dedicated to medical-grade films and materials. Based on primary research, the company's countries served and distribution networks can be estimated as broad across the major global regions, based on its 46 active production sites and 13 research facilities operating across the EMLA, NA, and APAC geographic markets.

SUPPORTING POINTS

- Covestro expanded its production capacities for thermoplastic polyurethane (TPU) films under the Platilon® range, which are utilized in medical wound dressings, at its Epurex Films center of excellence in Bomlitz, Germany.
- The company launched the localized regional production of its Desmopan® Rx medical-grade thermoplastic polyurethanes (TPU) at its newly qualified facility in Changhua, Taiwan, marking its second global site capable of medical-grade TPU manufacturing alongside New Martinsville, North America.

R&D Investment & Innovation Pipeline — scored 91/100

> The rating rests on Covestro's established global network of R&D and application centers tailored to TPU advancement and the active deployment of newly developed commercial products for medical film applications. Based on primary research, the company's medical film R&D allocation share can be estimated as focused on niche segments, based on its global distribution of innovation center investments across diverse industrial sectors such as automotive, building, electronics, and health.

SUPPORTING POINTS

- Covestro announced in 2024 that it is investing around EUR 100 million in its global R&D infrastructure and assets over a three-year timeframe until 2025, which includes upgrading laboratories in Leverkusen, Germany, that drive application technology for medical applications.
- The company announced the launch of Dureflex MA5000 ahead of MD&M West 2026, which is an innovative thermoplastic polyurethane (TPU) film on a paper carrier engineered specifically for high-performance medical wearables, IV securement dressings, and continuous glucose monitor (CGM) overpatches.

Market Share & Brand Leadership — scored 100/100

> The rating rests on Covestro's established brand infrastructure, global production volumes exceeding 45 kilotons, and active capacity expansions for its Platilon brand. Based on primary research, the company's medical films percentage market share can be estimated as strong, in line with leading vendors, based on its status as one of the world's largest polymer manufacturers completely geared toward the global circular economy.

SUPPORTING POINTS

- The company produced over 45 kilotons of thermoplastic polyurethane (TPU) films globally in 2024, demonstrating its widespread presence across regional markets.
- In October 2023, the company significantly increased its production capacity for thermoplastic polyurethane (TPU) films at its Bomlitz site in Germany to strengthen its Platilon range of adhesive films.

Eastman Chemical Company Tritan ● Leaders X 92 Y 93 Overall 92

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 91/100

> The company's portfolio rating rests on its clear commercialization of specialized medical-grade copolyesters and copolyester elastomers suited for films and flexible packaging. Based on primary research, the company's broader film-grade medical portfolio can be estimated as focused on specialty copolyesters and cellulosics, based on its historical focus on Eastar rigid packaging and Tritan medical devices.

SUPPORTING POINTS

- Eastman Chemical Company commercializes its specialized Tritan MP100 copolyester, a medical-grade material specifically formulated for medical films, sheets, and packaging applications.
- Eastman Chemical Company offers the Ecdel elastomer, which is a plasticizer-free copolyester ether (COPE) elastomer utilized in medical and bioprocessing bags.

Film Performance & Functional Capabilities — scored 98/100

> The evaluation rests on the company's clear and verifiable technical specifications published in its official data sheets for specialized medical film-grade copolyesters (MP100 and DA001), which outline precise metrics for tensile strength, clarity, and moisture permeability. Based on primary research, the company's performance benchmarks can be estimated as high, in line with leading medical material suppliers, based on its specialty copolyester materials, such as Tritan, which are widely utilized across demanding healthcare packaging and clinical applications.

SUPPORTING POINTS

- The company's published technical data sheet for Eastman Tritan™ copolyester MP100 (formulated for medical film, sheet, and rigid medical packaging) reports an optical clarity of 1% haze and 92% total transmittance, a tensile strength at yield of 40–41 MPa, a tensile strength at break of 52–59 MPa, and a water vapor permeability rate of 4–11 g/m²·24h.
- The company's technical product catalog confirms that Eastman Tritan™ copolyester DA001 is an amorphous copolyester engineered to provide the explicit optical clarity, toughness, flexibility, and chemical resistance needed for medically extruded film and thermoformed packaging applications.

Regulatory Compliance & Biocompatibility — scored 91/100

> The rating rests on confirmed compliance statements and product specifications for Eastman's specialized copolyesters and elastomers verifying that they satisfy USP Class VI and ISO 10993 biocompatibility requirements for non-implantable medical packaging and film applications. Based on primary research, the company's regulatory approvals can be estimated as fast for most approvals, based on its comprehensive biocompatibility data, extensive testing history, and long-standing regulatory compliance protocols for medical-grade polymers.

SUPPORTING POINTS

- The company reports that its Eastar 6763 copolyester, which is utilized for film and sheet extrusion in medical packaging applications, complies with ISO 10993 and USP Class VI biocompatibility evaluation requirements.
- The company reports that its Eastman Ecdel elastomers, used for flexible medical packaging and films, meet selected FDA-Modified ISO 10993 and USP Class VI biological evaluation testing criteria for contact times of 30 days or less.

Customization & Co-Development Capability — scored 93/100

> The rating is based on verified joint development case studies with medical equipment clients and self-reported custom options for its technical film portfolio. Based on primary research, the company's custom formulation projects can be estimated as flexible and responsive to specialty requirements, based on its roll-to-roll diagnostics capabilities, custom tailored functionalities, and established global technology centers.

SUPPORTING POINTS

- Eastman Chemical Company collaborated with Swedish medical technology firm CYTO365 to develop the RondelO turn valve for multi-inlet infusion therapy, leveraging the technical team's application development support to optimize accurate molding and transparency using Tritan copolyester, as announced in January 2024.
- The company reports that its specialized Flexvue films line provides custom tailoring capabilities where films can be modified with distinct functionalities to match specific market application needs.

Manufacturing Quality & Consistency — scored 86/100

> The rating rests on Eastman Chemical Company's internal enforcement of current Good Manufacturing Practice (cGMP) guidelines and self-reported process consistency for its specialty medical films and polymer lines. Based on primary research, the company's manufacturing site quality performance can be estimated as aligned with standard international chemical protocols, based on its strict reliance on global Good Manufacturing Practices and corporate regulatory compliance.

SUPPORTING POINTS

- The company reports implementing strict manufacturing and quality guidelines through current Good Manufacturing Practice (cGMP) regulations across its polymer production processes to ensure raw material traceability, reliability, and repeat repeatability for medical packaging and device components.
- Eastman Chemical Company highlights that its Flexvue sputtered films provide run-to-run and cross-web consistency, offering optimized resistance and optical properties roll after roll for the medical industry.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 90/100

> We verified that the company reports macro-level growth within its Advanced Materials business unit, which handles specialty copolyesters like Tritan used across medical packaging and fluid devices. Based on primary research, the company's breakout revenue or growth rates can be estimated as growing steadily, based on its overall advanced materials sequential revenue growth and specialty plastics performance.

SUPPORTING POINTS

- Eastman Chemical Company reports that its Advanced Materials operating segment—which manages and markets specialty polymers, films, and plastics for medical and pharma applications—achieved a 5% increase in sales revenue in Q2 2026, driven by higher sales volumes and favorable foreign currency exchange rates.

Named Customers & Contract Wins — scored 95/100

> We verified Eastman Chemical Company's established material supply partnerships with prominent healthcare device and packaging manufacturers like Sotera Wireless and Tekni-Films. Based on primary research, the company's recent contract wins or customer names can be estimated as strong, in line with leading vendors, based on its established partnership with medical pack suppliers like Medipack for sterile barrier film applications.

SUPPORTING POINTS

- Tekni-Films selected Eastar 6763 copolyester from Eastman Chemical Company to develop rigid films and thermoformed trays for the medical and pharmaceutical packaging industry, as reported by Tekni-Films.
- Sotera Wireless, Inc. incorporated Eastman Chemical Company's Tritan copolyester into the lens, housing, and connectors of its wrist-worn ViSi Mobile System continuous vital signs monitoring device, as published by Healthcare Packaging.

Global Production & Supply Footprint — scored 91/100

> We verified that the company's core specialty medical-grade polymer production is anchored across two major integrated North American chemical manufacturing sites, alongside a network of 36 global manufacturing locations spanning 12 countries. Based on primary research, the company's medical film facilities outside primary hubs can be estimated as limited to a few strategic locations, based on its 36 global manufacturing facilities, including key specialized film and interlayer extrusion hubs in locations like Dresden and Ghent.

SUPPORTING POINTS

- Eastman Chemical Company operates a dedicated manufacturing facility exclusively for its Tritan copolyester line at its global headquarters in Kingsport, Tennessee, USA.
- The company expanded its specialty copolyester manufacturing footprint to a second North American production site in Columbia, South Carolina, USA.

R&D Investment & Innovation Pipeline — scored 94/100

> The rating rests on the company's clear and verifiable capability in developing distinct medical-grade copolyesters (such as Tritan MP100, MP200, and DA001) targeted directly at the medical film and extruded sheet landscape. Based on primary research, the company's isolated R&D spending or patent pipelines can be estimated as high, based on its decades of experience in healthcare resin innovation and continuous patent filings for plasticized copolyesters.

SUPPORTING POINTS

- Eastman Chemical Company developed and commercialized specialized medical-grade copolyester resins formulated specifically for medical film, sheet, and packaging applications, including the amorphous Tritan MP100 and MP200 copolyesters.
- The company expanded its specialized medical film pipeline by introducing Eastman Tritan DA001 copolyester, an amorphous material engineered explicitly for medically extruded film applications meeting ISO 10993 and USP Class VI biocompatibility requirements.

Market Share & Brand Leadership — scored 94/100

> We verified the company's strong brand leadership and commercial footprint through its documented supply partnerships with major healthcare packaging manufacturers. Based on primary research, the company's market share percentage or segment revenue can be estimated as strong and growing steadily, based on its total global sales revenue of over $9 billion and prominent position within the medical polymer landscape.

SUPPORTING POINTS

- In November 2025, Switzerland-based packaging provider Medipack integrated the company's Eastar 6763 Renew copolyester into its MEDPETG 6763 RN thermoforming film to manufacture sustainable medical packaging, according to an Eastman Chemical Company Press Release.

Wacker Chemie AG SILPURAN ● Leaders X 92 Y 92 Overall 92

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 83/100

> We verified that the company's confirmed commercialized capabilities for this market are highly concentrated and specialized within silicone-based polymers and pre-formed silicone films. Based on primary research, the company's other medical-grade polymer type commercialization can be estimated as limited to a few markets, based on its core concentration on high-precision crosslinked silicone rubber films and silicone-based additives rather than standalone non-silicone thermoplastic medical film lines.

SUPPORTING POINTS

- The company reports the commercialization of its SILPURAN Film 2030, a biocompatible, cross-linked silicone rubber film specifically designed for breathable, atraumatic wound dressings and gas-permeable membranes in medical devices.
- The company commercializes the SILPURAN product line, which includes liquid silicone rubber (LSR) and high consistency rubber (HCR) polymers used to manufacture medical components, alongside platinum-catalyzed silicone gel adhesives (such as SILPURAN 2114 and 2124) tailored for medical lamination and wearable devices.

Film Performance & Functional Capabilities — scored 93/100

> The rating rests on extensive confirmed data sheets outlining technical performance capabilities for the company's dedicated medical silicone films. Based on primary research, the company's comparative benchmark evaluations can be estimated as strong, in line with leading vendors, based on its published comparative transdermal drug delivery studies showing above-average active ingredient release profiles against commercial benchmarks.

SUPPORTING POINTS

- The company reports that its SILPURAN Film 2030, a biocompatible platinum-cured crosslinked silicone rubber film designed for medical applications, is produced under class 8 cleanroom conditions to restrict thickness deviations to no more than plus or minus 5 percent across the web.
- According to the published product specifications for ELASTOSIL Film 2030 (sharing physical specifications with the medical equivalent), the technical capabilities include a tensile strength of 5.5 N/mm², 450% elongation, low/high temperature flexibility ranging from -40 °C to +180 °C, high transparency, and high gas and water-vapor permeability.

Regulatory Compliance & Biocompatibility — scored 98/100

> The rating rests on clear, verifiable proof of compliance testing for medical-grade silicone series under both ISO 10993 and USP Class VI frameworks across various application bounds. Based on primary research, the company's material batch verification records can be estimated as fast for most approvals, based on its standard provision of technical documentation and cleanroom manufacturing compliance under strict quality assurance frameworks.

SUPPORTING POINTS

- The company reports that its SILPURAN 4200 medical-grade moisture-curing silicone adhesive formulation passed ISO 10993 certifications for cytotoxicity, sensitization, and pyrogenicity, alongside USP Class VI certification for systemic toxicity, intracutaneous toxicity, and 5-day implantation.
- The company reports that its SILPURAN 2117 two-part biocompatible soft skin adhesive gel, manufactured within an ISO Class 8 cleanroom for direct-skin wound care applications, has successfully passed ISO 10993 and USP Class VI compliance testing.

Customization & Co-Development Capability — scored 98/100

> The rating rests on the company's verified acquisition of a dedicated medical silicone prototyping and contract manufacturing platform to provide tailored processing recommendations and custom-coated components to medical device designers. Based on primary research, the company's named joint development case studies can be estimated as growing steadily, based on its collaborative partnerships like the direct coating joint technology development project with Freudenberg Performance Materials.

SUPPORTING POINTS

- Wacker Chemie AG announced the acquisition of BioMed Sciences, Inc.'s manufacturing assets and technology in 2024 to establish a prototyping and contract manufacturing platform under Wacker Silicone Medical Devices (WSMI), allowing the company to directly support medical film coaters, converters, and device designers with custom stick-to-skin applications.
- The company reports that its standard two-component medical silicone skin adhesives, such as the SILPURAN 2100 A/B gel system, allow customers to customize mechanical properties and tackiness directly during processing by adjusting the component mixing ratios to alter the final vulcanisate hardness.

Manufacturing Quality & Consistency — scored 88/100

> The evaluation rests on the company's internal clean operations standard and clean room packaging protocols for its medical silicone range. Based on primary research, the company's manufacturing site certifications can be estimated as strong, in line with leading vendors, based on its dedicated SILPURAN medical technology production environments and standard regulatory master file alignments expected of prime global chemical producers.

SUPPORTING POINTS

- The company reports that all SILPURAN and ELASTOSIL LSR and RTV products are produced according to the WACKER CLEAN OPERATIONS standard, utilizing selected mixers, 50 µm filtration, and an air-controlled filling environment to meet medical cleanliness needs.
- The company reports that its SILPURAN medical-grade silicone rubber products are filled and packaged in Class 8 clean rooms to guarantee high purity levels and comply with ISO 10993 and USP Class VI biocompatibility standards.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 69/100

> We verified that the company discloses overall corporate and division-level revenue for its chemical segments (Silicones and Polymers), which serve medical technology fields. Based on primary research, the company's breakout revenue for medical film raw materials can be estimated as growing steadily, based on its over 5 billion euro annual sales across various key industries including medical technology.

SUPPORTING POINTS

- The company reports in its Annual Report 2025 that its overall group sales for fiscal 2025 reached approximately €5.49 billion, with the Silicones division—which encompasses medical technology applications—generating annual sales of €2.73 billion, representing a 3 percent decline from €2.81 billion in 2024.

Named Customers & Contract Wins — scored 100/100

> We verified that the company has established long-term material supply agreements and partnerships with notable medical device and wound care companies like Mölnlycke Health Care and Trio Healthcare. Based on primary research, the company's newer contract manufacturing unit metrics can be estimated as limited to a few markets, based on its acquisition of bio med sciences assets and establishment of a new entity in allentown pennsylvania.

SUPPORTING POINTS

- Mölnlycke Health Care honored Wacker Chemie AG with a supplier award in the Long-Term Relationship category, celebrating their steadfast partnership supplying medical silicones since the late 1990s.
- Trio Healthcare uses Wacker Chemie AG’s SILPURAN silicone gel adhesives to attach its ostomy care systems to skin, ensuring leak-free performance and minimizing irritation.

Global Production & Supply Footprint — scored 98/100

> The company has demonstrated a robust, cross-continental manufacturing base for specialty silicone materials including medical-grade elastomers, supported by 27 global production sites distributed across Europe, the Americas, and Asia. Based on primary research, the company's dedicated production capacity for backing polymers can be estimated as minor compared to standard medical components, based on its extensive portfolio of standard liquid silicone rubbers for medical wearables and tubing.

SUPPORTING POINTS

- Wacker Chemie AG officially opened a new specialty silicone production facility in Karlovy Vary, Czech Republic, in June 2026, which manufactures room-temperature-curing silicones and solid silicone rubber targeting growth sectors including medical technology.
- Wacker Chemie AG completed a major capacity expansion program across its global network, expanding liquid silicone rubber (LSR) capacity at its Adrian, Michigan plant, and introducing high-consistency rubber (HCR) and specialty silicone manufacturing at sites in Panagarh (India), Plzeň (Czech Republic), and Tsukuba (Japan).

R&D Investment & Innovation Pipeline — scored 91/100

> We verified that the company actively develops medical-grade silicone gels and protective films, backed by dedicated facility trials and recent eco-labeled material extensions. Based on primary research, the company's segment global research and development spending can be estimated as minor portion of total group innovation investments, based on its over 200 million euros in total group research and development expenses.

SUPPORTING POINTS

- Wacker Chemie AG announced the launch of SILPURAN eco 2114 at the COMPAMED 2025 trade show, a biomethanol-based silicone gel adhesive designed for the fixation of medical wound dressings, wearables, and monitoring patches.
- Wacker Chemie AG presented research from its Healthcare Competence Center in Ann Arbor (USA) using Franz diffusion cell methodology to evaluate SILPURAN silicone gel adhesives as a stable, biocompatible matrix for controlled drug delivery in transdermal patches.

Market Share & Brand Leadership — scored 100/100

> The rating rests on the company's verified position as the world's second-largest silicone manufacturer and its strategic integration into the medical coating and film conversion space via the acquisition of Bio Med Sciences' assets. Based on primary research, the company's mathematical sector market share can be estimated as strong, in line with leading vendors, based on its position as a global specialty chemicals supplier present in all relevant markets.

SUPPORTING POINTS

- The company reports being the undisputed number-two player in the global market for silicones and leverages this scale to supply its specialized SILPURAN FILM 2030 series directly to the healthcare industry.
- In May 2024, Wacker Chemie completed the acquisition of Bio Med Sciences' assets and technology to transition into a vertically integrated solution provider of silicone-coated healthcare patches and medical device adhesive coatings.

Kraton Corporation (subsidiary of DL Chemical) Kraton ● Leaders X 90 Y 84 Overall 87

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 90/100

> We verified that the company's confirmed capabilities rest entirely on its proprietary styrenic block copolymers (SBCs)—specifically hydrogenated styrenic block copolymers (HSBC) such as SEBS and sulfonated block copolymers—engineered for medical compounding and barrier films. Based on primary research, the company's commercialization of medical film materials can be estimated as absent from its core offering, based on its focused production of styrenic block copolymers and hydrogenated styrenic block copolymers for medical applications.

SUPPORTING POINTS

- The company reports that its Kraton G1645 polymer is a hydrogenated styrenic block copolymer (HSBC) designed specifically for single or multilayer medical IV film applications, engineered to be compounded with polypropylene (PP).
- The company reports that its Nexar polymer technology is a specialty sulfonated block copolymer family commercialized as an anti-fog film and coating solution for medical personal protective equipment (PPE).

Film Performance & Functional Capabilities — scored 93/100

> The rating rests on confirmed technical data sheets for the company's Kraton G and Kraton D polymer families highlighting specific tensile strengths, elongation boundaries, and steam sterilization parameters optimized for medical-grade films. Based on primary research, the company's clinical performance benchmark sheets can be estimated as highly competitive, matching leading alternatives, based on its extensive ISO 112 and USP Class VI regulatory compliance validation.

SUPPORTING POINTS

- The company reports in its technical data sheet for Kraton G1645 polymer—a hydrogenated styrenic block copolymer (HSBC) using enhanced rubber segment (ERS) midblock technology—that the material is specifically designed for single or multilayer medical IV film applications to deliver steam sterilization capability, heat seal strength, and impact resistance.
- According to Kraton's performance product guide, its D1114 styrene-isoprene-styrene (SIS) grade used in elastic films demonstrates a tensile strength of 16 MPa in the machine direction (MD) and 17 MPa in the transverse direction (TD), along with an elongation at break of 1145% (MD) and 1205% (TD).

Regulatory Compliance & Biocompatibility — scored 91/100

> The rating rests on the company's clear self-reported compliance for its primary medical-grade polymers, which are certified under USP Class VI and ISO 10993 testing parameters. Based on primary research, the company's clinical trial case study verification can be estimated as regulatory and standard test benchmark compliant, based on its published adherence to standard international biocompatibility guidelines including United States Pharmacopeia Class 6 and ISO 31 requirements.

SUPPORTING POINTS

- The company reports that its specialized hydrogenated styrenic block copolymer (HSBC) grades, specifically Kraton G1643 and Kraton G1645 used in flexible film extrusion and polyolefin blending, pass all ISO 10993 biological toxicity testing and maintain compliance with USP Class VI requirements, as detailed in Kraton Polymers -.
- The company reports that its Kraton Quality Management System (KQMS) maintains dedicated medical regulatory certifications, including direct Drug Master File (DMF) listings, pharmacopeia compliance, and certified testing per FDA requirements, according to the Kraton Corporation Medical Applications Guide.

Customization & Co-Development Capability — scored 93/100

> The rating rests on confirmed multi-layer film co-development projects and a structured market development framework for piloting customized medical formulations with clients. Based on primary research, the company's custom thickness drug compatibility metrics can be estimated as customer specialized and application specific, based on its application specifications providing generic blending guidance with polypropylene for variable multi-layer film compounding structures.

SUPPORTING POINTS

- The company reports a partnership with medical film manufacturers to co-develop a polyolefin-based intravenous (IV) film architecture utilizing the Kraton G1645 polymer combined with polypropylene in cast and blown film multi-layer IV bags.
- The company utilizes a distinct "Market Development" (MD) protocol where advanced polymers in the custom design and evaluation cycle are designated with an MD prefix before commercialization, allowing medical customers to test tailored properties like clean, oil-free thermoplastic styrenic formulations.

Manufacturing Quality & Consistency — scored 84/100

> The rating rests on the company's self-reported implementation of a global quality management system alongside standard medical polymer certifications. Based on primary research, the company's manufacturing facility manufacturing line quality audits can be estimated as robust regulatory standard compliance, based on its global supply chain operations distributing highly-engineered specialty polymers to diverse medical and industrial customers in over 70 countries.

SUPPORTING POINTS

- The company reports implementing the Kraton Quality Management System (KQMS) to maintain harmonized processes, ensuring its medical polymers meet regulatory compliance including USP Class VI, ISO 10993 testing, and active Drug Master File (DMF) listings.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 78/100

> We verified that Kraton Corporation reports its consolidated finances across its primary Polymer and Chemical segments, but does not disclose breakout figures for healthcare or medical film materials. Based on primary research, the company's medical film revenue growth can be estimated as growing steadily, based on its global polymer operations across over 70 countries.

SUPPORTING POINTS

- The company reported that the net revenue impact from the sale of its compounding business—which directed compound sales primarily into personal care, protective film, and general compounds for consumer, medical, and automotive applications—was less than two percent of consolidated revenue for fiscal year 2014.

Named Customers & Contract Wins — scored 70/100

> We verified that the company's publicly available documentation lists target applications such as medical films and IV bag materials without naming specific commercial customer wins or supply contracts. Based on primary research, the company's medical film client names can be estimated as broad across major global medical device manufacturers, based on its targeted PVC-replacement polymer solutions compliance with ISO and USP Class VI standards.

SUPPORTING POINTS

- The company reports providing its hydrogenated styrenic block copolymers (SBC) and enhanced rubber segment (ERS) polymers for medical bags, single/multi-layer medical films, and medical tubing applications globally, but does not disclose specific converter or device brand customer names.

Global Production & Supply Footprint — scored 91/100

> We verified that the company has established manufacturing sites across six countries and serves customers in over 70 countries globally with its specialty polymers portfolio. Based on primary research, the company's medical production capacities can be estimated as strong in line with leading global specialty polymer facilities, based on its extensive global manufacturing network backed by DL Chemical.

SUPPORTING POINTS

- Kraton Corporation reports operating manufacturing facilities across six countries globally to support its polymer product availability, which includes sites in Paulinia (Brazil), Berre (France), Wesseling (Germany), Pernis (The Netherlands), Kashima (Japan joint venture), and its flagship US plant in Belpre, Ohio.
- The company announced a 30% capacity expansion project scheduled to begin in the third quarter of 2024 at its hydrogenated styrenic block copolymer (HSBC) joint venture plant, Kraton Formosa Polymer Corporation, located in Mailiao, Taiwan, to serve its growing customer base in Asia.

R&D Investment & Innovation Pipeline — scored 91/100

> We verified that the company holds recent active patents for protective laminate structures and reports active collaborations on recyclable, multi-layer polyolefin medical film architectures. Based on primary research, the company's medical film R&D investment can be estimated as scaling significantly, based on its heavy R&D investment focus on sustainable innovation and advanced medical material technology.

SUPPORTING POINTS

- Justia Patents lists Patent number 12584052, assigned to Kraton Corporation and granted on March 24, 2026, which covers a laminate structure featuring a support layer and a self-sterilizing/antimicrobial component.
- Kraton Corporation reports a recent co-development project with medical film manufacturers to enable a polyolefin-based IV film architecture that blends its Kraton G1645 styrenic block copolymer with polypropylene to replace PVC structures.

Market Share & Brand Leadership — scored 92/100

> The rating rests on the company's confirmed historical global brand leadership in styrenic block copolymer technologies and its established industry status as a specialized polymer supplier for medical films and non-PVC IV bag components. Based on primary research, the company's post-merger medical market share can be estimated as strong and leading, based on its established global footprint as a primary specialty polymer supplier under DL Chemical's financial strength.

SUPPORTING POINTS

- The company reports in historical regulatory filings that it maintains the number one leading market position globally for its core styrenic block copolymer segments, explicitly listing medical and PVC-replacement multi-layered films within its key packaging and films application portfolio.

Poly-Med, Inc. Poly-Med ● Leaders X 87 Y 81 Overall 84

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 78/100

> We verified that the company's confirmed film polymer capabilities rest entirely on its specialized portfolio of proprietary bioresorbable, absorbable polyester-based polymers (such as PLA, PGA, PDO, PCL, and TMC blends) rather than commodity or non-absorbable film polymers. Based on primary research, the company's a commercial medical-grade portfolio of standard medical film polymers can be estimated as limited to a few markets, based on its bioresorbable polyester polymers and implantable device focus.

SUPPORTING POINTS

- The company reports offering its proprietary Max-Prene (PGLA), Glycoprene (PGA/TMC/caprolactone), Strataprene, Dioxaprene (PDO), and Lactoprene (PLLA-based) bioresorbable polyester-based polymer series, which can be melt-extruded or solvent-processed into degradable barrier films of 70–300 micron thickness for direct tissue contact.

Film Performance & Functional Capabilities — scored 93/100

> The assessment rests on peer-reviewed engineering data evaluating the tensile strength, elasticity, and hydrolytic degradation timelines of extruded and solvent-cast bioresorbable film layers. Based on primary research, the company's standard commercial data sheets detailing breathability or water vapor transmission rates can be estimated as limited to a few markets, based on its implantable barrier films and surgical device applications.

SUPPORTING POINTS

- A study presented at the Society for Biomaterials annual meeting showed that films created from Poly-Med's Glycoprene RD7—a 25:20:55 poly(TMC-co-caprolactone-co-glycolide) copolymer—extruded to a thickness of 25 µm, were evaluated under ASTM F1635-11 and ASTM D882-10 protocols to measure mass loss and tensile strength retention during in vitro degradation in a 7.4 pH phosphate buffer at 37°C.
- Technical evaluations by Poly-Med demonstrated the development of strong, flexible, and highly elastic films prepared from a high molecular weight polyaxial co-polyester dissolved in acetone via solvent casting, specifically designed to overcome the stiffness and noncompliant properties of traditional amorphous 70:30 poly(L-lactide-co-D,L-Lactide) materials.

Regulatory Compliance & Biocompatibility — scored 85/100

> We verified that the company maintains an FDA registration and aligns its facility operations with 21 CFR Part 820 quality management system frameworks. Based on primary research, the company's standalone certifications for medical film applications can be estimated as strong, in line with leading vendors, based on its ISO 102 certified quality management and cGMP-compliant manufacturing.

SUPPORTING POINTS

- The company reports operating an FDA-registered testing lab and a manufacturing facility maintained in compliance with the FDA 21 CFR Part 820 Quality System Regulation.

Customization & Co-Development Capability — scored 93/100

> The rating rests on confirmed self-reported evidence of custom material engineering, thickness control, and specialized film-processing methods targeting the implantable and bioresorbable medical film segment. Based on primary research, the company's joint-development case studies or client names can be estimated as limited to a few markets, based on its custom biomedical device design and original equipment manufacturer partnership model.

SUPPORTING POINTS

- Poly-Med, Inc. reports that it designs and manufactures custom bioresorbable polymers, adjusting the chemical composition and block copolymer structure—such as its Strataprene series designed for films—to achieve specific mechanical stiffness, elasticity, and degradation timelines tailored to client requirements.
- The company reports that its custom film extrusion capabilities support both solvent processing for heat-sensitive drug-delivery formulations and flat-die melt extrusion, allowing film thickness customization typically ranging from 0.05 mm to 0.5 mm (or 70 to 300 microns) on roles with or without backing material.

Manufacturing Quality & Consistency — scored 88/100

> The evaluation rests on confirmed internal QMS compliance frameworks including official ISO 13485:2016 certification and FDA registration for its custom bioresorbable polymer manufacturing facility. Based on primary research, the company's customer performance metrics or quantified lot-to-lot defect rates can be estimated as strong, in line with leading vendors, based on its vertically integrated single-source provider capabilities from polymerization to extrusion.

SUPPORTING POINTS

- The company reports that its design, development, and manufacturing operations for bioresorbable polymers, fibers, sutures, and medical textiles at its Anderson, South Carolina facilities are certified to meet ISO 13485:2016 standards for quality management.
- The company reports operating its production facility in alignment with US FDA Quality System Regulations outlined under 21 CFR Part 820, utilizing an in-house analytical testing laboratory to validate critical testing protocols against standard benchmarks such as ASTM, ISO, and ANSI to support repeatability and lot uniformity.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 69/100

> We verified that Poly-Med, Inc. supplies bioresorbable polymers for processing formats including film extrusion, but the firm does not publish segmented financial reports. Based on primary research, the company's revenue from polymer sales or growth trajectory can be estimated as growing steadily, based on its 30,000 square foot facility expansion and doubled manufacturing capabilities.

SUPPORTING POINTS

- The company has an estimated overall annual revenue of 18.2 million dollars, according to a market overview.

Named Customers & Contract Wins — scored 84/100

> The rating rests on confirmed development partnerships with Novus Scientific and Formlabs involving custom bioresorbable constructs and SLA 3D printing equipment deployment. Based on primary research, the company's commercial-scale supply agreements or contract wins can be estimated as strong, in line with leading vendors, based on its 15 product marketplace releases and custom bioresorbable extrusion offerings.

SUPPORTING POINTS

- In a contract development partnership announced prior to public litigation filings, Poly-Med, Inc. developed a specialized surgical mesh under an exclusive agreement for the medical device firm Novus Scientific for use in hernia-repair products.
- In October 2025, Formlabs announced that Poly-Med, Inc. utilizes its Open Material Mode (OMM) on Formlabs SLA 3D printers to test and develop novel biomaterials, including Poly-Med's proprietary Photoset 023 resin.

Global Production & Supply Footprint — scored 88/100

> We verified that the company's manufacturing presence rests entirely on its centralized footprint of three production facilities in South Carolina, USA, which handle custom synthesis and specialized film extrusion of bioresorbable polymers. Based on primary research, the company's global distribution network or countries served can be estimated as broad across international life sciences communities, based on its global partnerships fostered via scbio and 25 years of industry innovation.

SUPPORTING POINTS

- According to the official Poly-Med Locations Overview, the company operates three distinct manufacturing facilities in South Carolina, USA, encompassing 121,000 square feet across Anderson (a 27,000 sq ft corporate headquarters and Biomaterial Center of Excellence), Piedmont (an Extrusion Center of Excellence housing film extrusion), and Liberty.
- The US FDA registers exactly two specific device facility sites for the company—one in Anderson and one in Piedmont—as independent regulatory records verified by BioBridgeX's Poly-Med Profile.

R&D Investment & Innovation Pipeline — scored 91/100

> The rating rests on the company's active intellectual property pipeline, including its milestone of submitting over 100 patent applications to the USPTO and its documented material innovations in bioresorbable block copolymers. Based on primary research, the company's annual r&d spending figures can be estimated as high, based on its 25 years of bioresorbable material innovation and comprehensive design services.

SUPPORTING POINTS

- The company reports the development and optimization of its Strataprene elastomeric polymer series, designed for tissue spacer, coating, and graft applications, leveraging its vertically integrated melt-extrusion and solvent processing film lines.
- In 2025, Poly-Med, Inc. was granted US Patent 12,398,236 for absorbable block copolymers demonstrating improved thermal stability, molecular weight consistency, and retention of inherent viscosity following melt extrusion.

Market Share & Brand Leadership — scored 75/100

> We verified that the company offers specific bioresorbable polymer film extrusion capabilities. Based on primary research, the company's annual medical film revenues or production volumes can be estimated as increasing significantly, based on its custom melt extrusion and solvent processing capabilities for films up to 30 centimeters wide.

SUPPORTING POINTS

- The company reports that it provides custom extrusion of its portfolio of bioresorbable polymers for the development and manufacturing of bioresorbable films, capable of extruding bioresorbable films with thicknesses ranging from 70 to 300 microns and widths up to 12.7 centimeters.

Xuchuan Chemical (Suzhou) Co., Ltd. Xuchuan ● Challengers X 80 Y 85 Overall 82

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 81/100

> We verified that the company's verified medical film capability rests exclusively on its self-reported thermoplastic polyurethane resin line. Based on primary research, the company's broader range of commercialized medical-grade film polymers can be estimated as limited to polyurethane variants, based on its specialization in thermoplastic polyurethane resin and polyester polyols for films and sheets.

SUPPORTING POINTS

- The company reports commercializing its thermoplastic polyurethane (TPU) plastic granules under Model Number 3285, which are engineered for application in films and sheets for medical supplies, as documented on the official Xuchuan Chemical Product Page.

Film Performance & Functional Capabilities — scored 86/100

> The rating rests on the company's verified commercialization of a dedicated medical film polymer granule under Model No. 3285. Based on primary research, the company's mechanical and barrier performance benchmarks can be estimated as strong in elasticity and abrasion resistance, based on its published properties emphasizing high mechanical strength, wear resistance, and oil resistance for thermoplastic polyurethanes.

SUPPORTING POINTS

- The company reports the production of Thermoplastic Polyurethane plastic granules under Model No. 3285, which are designated explicitly for application in films and sheets for medical supplies.

Regulatory Compliance & Biocompatibility — scored 79/100

> We verified that the company highlights its research, development, and high-volume production capabilities for thermoplastic polyurethanes across civil, automotive, and medical fields. Based on primary research, the company's biocompatibility and regulatory compliance capabilities can be estimated as safe for basic non-allergenic contact, based on its environmental compliance testing indicating that its thermoplastic polyurethanes do not contain heavy metals.

SUPPORTING POINTS

- Xuchuan Chemical (Suzhou) Co., Ltd. maintains a corporate video profile demonstrating its technical alignment for "Applications of TPU in Food Packaging and Medical Devices."

Customization & Co-Development Capability — scored 81/100

> We verified that the company's customization capability relies entirely on generalized marketing materials for its broader polyol and standard TPU product lines rather than medical film applications. Based on primary research, the company's custom formulation and drug-compatibility project scope can be estimated as focused primarily on raw material optimization and processing adjustments, based on its documented focus on raw material modification and formula tuning to expand standard thermoplastic polyurethane film applications.

SUPPORTING POINTS

- The company reports offering molecular structure design and customization of polyester polyol product characteristics for different application scenarios to ensure perfect synergy with isocyanates, as promoted at the K Show 2025 exhibition.

Manufacturing Quality & Consistency — scored 73/100

> We verified that the company highlights the production of medical-grade TPU resins and films for medical equipment applications, but no specific independent quality verifications or case studies were identified. Based on primary research, the company's medical-line quality and processing management capabilities can be estimated as robust for mass industrial extrusion, based on its scale as a national high-tech enterprise running 4 modern production bases with over 600,000 tons of annual capacity.

SUPPORTING POINTS

- The company reports offering high-quality medical-grade TPU materials and TPU film explicitly targeted for medical supplies and medical equipment application fields.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 100/100

> We verified that the company's financial disclosure is limited to its corporate-wide revenue of US$1.7 billion, which spans all of its primary civilian and industrial polyurethane sectors. Based on primary research, the company's medical segment revenue growth can be estimated as growing steadily, based on its annual capacity of 1.15 million tons and 2022 turnover of $1.7 billion.

SUPPORTING POINTS

- Xuchuan Chemical reports a company-wide total turnover of US$1.7 billion for the year 2022, which includes operations across its main polyurethane lines covering civilian, industrial, automotive, and medical equipment application fields.

Named Customers & Contract Wins — scored 65/100

> We verified that Xuchuan Chemical (Suzhou) Co., Ltd. promotes thermoplastic polyurethane (TPU) resins and films for medical devices. Based on primary research, the company's named customers or medical supply contracts can be estimated as extensive and globally diversified, based on its 5 production bases, annual capacity of 1.15 million tons, and export footprint spanning over 80 countries.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 85/100

> We verified the company’s broad operational infrastructure consists of multiple Chinese manufacturing plants supporting bulk chemical supply lines globally, yet its primary focus is on standard footwear, industrial, and packaging materials. Based on primary research, the company's isolated medical cleanroom footprint can be estimated as integrated within main infrastructure, based on its 5 modern production bases and 10 production lines.

SUPPORTING POINTS

- Xuchuan Chemical (Suzhou) Co., Ltd. reports operating 5 centralized production bases located in Jiangsu, Zhejiang, and Fujian provinces in China, reaching a total annual chemical production capacity of 1.15 million tons across its product lines.
- Xuchuan Chemical (Suzhou) Co., Ltd. reports that its general polyurethane products are exported to overseas business channels covering over 60 countries.

R&D Investment & Innovation Pipeline — scored 85/100

> The rating rests on the company's confirmed high-level investment metric of allocating over 3% of its turnover into its R&D activities and its broader polyurethane patent portfolio. Based on primary research, the company's medical TPU R&D expenditure can be estimated as strong, in line with leading vendors, based on its status as a national high-tech enterprise with an active R&D department.

SUPPORTING POINTS

- The company reports that it invests more than 3% of its annual sales revenue into research and development, maintaining a full-time R&D team consisting of more than 165 R&D staff as of 2026.
- The company reports that it has applied for a total of 30 invention patents, of which 17 have been authorized to secure its product formulations.

Market Share & Brand Leadership — scored 90/100

> The rating rests on the company's confirmed domestic market dominance and scale in broader polyurethane resins, as well as its self-reported global coverage spanning over 60 countries. Based on primary research, the company's medical film market share can be estimated as modest compared to core segments, based on its dominant product focus on footwear resins and polyester polyols.

SUPPORTING POINTS

- The company reports having 5 production bases in Jiangsu, Zhejiang, and Fujian, a total annual production capacity of 1.15 million tons, and claims the top rank in domestic polyurethane resin market share within China.

Elastron Kimya San. ve Tic. A.Ş. Elastron V Challengers X 82 Y 79 Overall 80

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 73/100

> We verified that the company's verified medical-grade polymer portfolio is limited to specialty polyolefin-based TPE and TPV compounds (SEBS and EPDM/PP blends) rather than core medical film polymer categories. Based on primary research, the company's commercialized medical-grade portfolios can be estimated as limited to specialized polyolefin-bondable formulations, based on its focus on EPDM and PP based thermoplastic vulcanizates bondable only to polyethylene polypropylene and ethylene vinyl acetate.

SUPPORTING POINTS

- The company reports the commercialization of its Elastron V series, which includes EPDM/PP-based thermoplastic vulcanizate (TPV) medical compounds such as the Elastron V P.V101.A60.B grade designed for medical applications, as detailed in product literature.
- The company reports the commercialization of its Elastron G series, featuring styrene-ethylene-butylene-styrene (SEBS) block copolymer-based thermoplastic elastomer (TPE) compounds like the medical-grade Elastron G P.G401.A55.N, as detailed in product documentation.

Film Performance & Functional Capabilities — scored 87/100

> The rating rests on the company's verified provision of medical-grade TPV compound specifications detailing processing adaptability and sterilization durability. Based on primary research, the company's functional thin film metrics can be estimated as typical of standard medical extrusion grades, based on its focus on medical device applications like tubes caps and gaskets rather than thin-film benchmarks.

SUPPORTING POINTS

- The company reports in its medical product selection documentation that its Elastron P.V101 compound series is formulated as an EPDM/PP-based thermoplastic vulcanizate (TPV) optimized for extrusion and injection moulding in medical environments.
- The company reports in its technical marketing guides that its medical elastomer grades successfully sustain functional performance after undergoing multiple sterilization cycles with steam, ethylene oxide (EtO), and gamma irradiation.

Regulatory Compliance & Biocompatibility — scored 91/100

> The rating rests on the company's self-reported suite of basic pharmacopeia test alignments and raw material conformity certificates for its TPV grades. Based on primary research, the company's regulatory master file configurations can be estimated as aligned with global medical compliance standards, based on its testing according to ISO 31 and United States Pharmacopeia standards.

SUPPORTING POINTS

- Elastron Kimya reports that its medical-grade thermoplastic elastomer and thermoplastic vulcanizate (TPV) compounds achieve conformity with USP Class VI (USP <88> in vivo biocompatibility), USP <87> in vitro cytotoxicity, and ISO 10993-5 standards.
- The company's P.V101 medical TPV product line, including grades ranging from P.V101.A50.N to P.V101.A90.N, maintains documented raw material compliance under FDA 21 CFR regulations and EU Regulation No. 10/2011.

Customization & Co-Development Capability — scored 82/100

> The rating rests on the company's generalized corporate self-reporting of co-creation capabilities and tailor-made elastomer development. Based on primary research, the company's custom medical technical initiatives can be estimated as driven by localized co-development requirements, based on its offering of dedicated customer solutions in selected European countries.

SUPPORTING POINTS

- The company reports in its corporate strategy that co-creation is a core brand value, stating that it specializes in crafting tailor-made TPE and TPV solutions designed to align with its customers' unique needs via a collaborative environment according to Elastron About Us.

Manufacturing Quality & Consistency — scored 75/100

> We verified that the company's quality frameworks rely on automotive and generalized industrial quality management standards, but it lacks medical-specific manufacturing quality systems. Based on primary research, the company's production line compliance capability can be estimated as strong and aligned with international device frameworks, based on its established compliance with recognized medical standards and ISO quality protocols.

SUPPORTING POINTS

- The company holds active corporate quality management certifications including ISO 9001:2015 and IATF 16949:2016 for its manufacturing facilities, as verified by SGS.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 78/100

> We verified that Elastron Kimya operates facilities in Turkey and the USA to supply custom thermoplastic vulcanizate (TPV) and elastomer compounds to the healthcare sector. Based on primary research, the company's segmented financial revenue, film-grade material sales volumes, or historical growth trajectories for its can be estimated as growing steadily, based on its IMCD partnership spanning all European nations and India alongside its medical-grade G and V series TPE portfolios.

SUPPORTING POINTS

- The company reports its production and logistics setup spans an official manufacturing facility in Gebze, Turkey and a second facility in Gainesville, Georgia, USA, which serve its global markets with specialty TPE and TPV solutions across automotive, consumer, industrial, and medical segments as documented on the Elastron corporate overview.

Named Customers & Contract Wins — scored 70/100

> The evaluation rests on a confirmed commercial distribution agreement spanning Europe and India. Based on primary research, the company's any named medical device or wound care manufacturers utilizing its materials or any publicly announced supply can be estimated as limited to a few markets, based on its publicly stated commercial supply track record for medical applications like syringe tips and its multi-country logistics network.

SUPPORTING POINTS

- Elastron Kimya San. ve Tic. A.Ş. entered a distribution partnership agreement with IMCD Group to distribute its medical range of TPE-S and TPE-V compounds across European countries and India.

Global Production & Supply Footprint — scored 91/100

> The evaluation is based on the company's verified footprint of two production plants across Turkey and the US, a dedicated compounding line for medical grades at its primary site, and a distribution reach exceeding 50 countries. Based on primary research, the company's annual production capacity allocated exclusively to medical-grade TPV compounds out of its total 25,000 can be estimated as small compared to industrial volumes, based on its primary operational facilities located in Gebze and Gainesville together with its extensive focus across automotive and construction segments.

SUPPORTING POINTS

- The company reports operating two primary production facilities, including its main factory and headquarters in Gebze, Kocaeli, Turkey, and a secondary manufacturing plant located in Gainesville, Georgia, USA, to serve the North American market.
- Elastron reports that its global service and distribution network exports compounds to over 75 countries across 6 continents, supported by a network of more than 70 distribution partners and international sales offices in Germany, China, and Japan.

R&D Investment & Innovation Pipeline — scored 75/100

> We verified that the company maintains an active R&D unit specializing in custom-tailored medical-grade TPE and TPV compound engineering for devices and caps. Based on primary research, the company's R&D spending figures, patent pipelines, or unique product developments dedicated to the medical films segment can be estimated as strong, in line with leading vendors, based on its dedicated focus on specialized medical grades passing United States Pharmacopeia Class 6 biocompatibility and cytotoxicity standards.

SUPPORTING POINTS

- The company reports operating a dedicated Elastron R&D center that develops customer-oriented and sustainable thermoplastic elastomer solutions, backed by a technical team possessing a combined engineering experience of over 75 years.

Market Share & Brand Leadership — scored 82/100

> We verified that the company actively competes in the medical-grade thermoplastic elastomers sector through its specialized medical V and G series TPE portfolios. Based on primary research, the company's volume production figures, annual revenues within the healthcare vertical, or global market share calculations can be estimated as emerging steadily, based on its distribution footprint touching over 75 countries globally via dedicated distribution partnerships.

SUPPORTING POINTS

- Elastron Kimya San. ve Tic. A.Ş. entered into a distribution agreement with Chase Plastics to expand the reach of its Elastron V (thermoplastic vulcanizate) product lines across automotive, medical, and industrial sectors.

Huntsman Corporation ● Challengers

X 79 Y 80 Overall 80

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 83/100

> We verified that the company's commercialized offerings for film extrusion focus entirely on thermoplastic polyurethanes (TPU). Based on primary research, the company's polymer family availability can be estimated as limited to polyurethane variants, based on its exclusive focus on thermoplastic polyurethane portfolios like IROGRAN and KRYSTALGRAN.

SUPPORTING POINTS

- The company's official digital product selector details a thermoplastic polyurethane (TPU) portfolio including the IROGRAN brand, spanning aliphatic, aromatic, caprolactone-, ester-, and ether-based chemistries for film and sheet extrusion applications.

Film Performance & Functional Capabilities — scored 91/100

> We verified that the company publishes extensive physical data sheets outlining mechanical parameters such as tensile strength, elongation, and tear resistance for its core IROGRAN extrusion polymer lines suitable for films. Based on primary research, the company's performance metrics disclosure can be estimated as general across industrial film sheets, based on its technical data sheets optimizing general extrusion and textile lamination without healthcare metrics.

SUPPORTING POINTS

- The company reports that its IROGRAN E, PS (ester-based), and IROGRAN P (ether-based) TPU polymer grades provide tunable breathability alongside high durability, puncture, abrasion, and tear resistance when used in barrier-type structures for containment films and sheet applications.
- A product data sheet published by Huntsman Corporation for IROGRAN A 92 P 4637, a polyether-based thermoplastic polyurethane used for film and sheet extrusion, establishes technical performance benchmarks including a tensile strength of 45 MPa, an elongation at break of 500%, and a tear strength of 90 N/mm under DIN and ISO norm testing.

Regulatory Compliance & Biocompatibility — scored 65/100

> We verified that Huntsman Corporation's public TPU documentation explicitly lists general compliance standards such as FDA and food contact approvals. Based on primary research, the company's biocompatibility data transparency can be estimated as available upon corporate request, based on its policy of requiring direct customer inquiries for specialized film series datasheets.

SUPPORTING POINTS

- The company's global TPU product selector lists certifications for standard industrial and consumer parameters including FDA, FCM (Food Contact Material), HFFR, NSF, and UL, but omits dedicated medical-grade bio-clearance options.

Customization & Co-Development Capability — scored 73/100

> We verified that the company offers medical-grade thermoplastic polyurethanes through application-focused engineering teams, but no documented joint development projects or custom formulation case studies were found within the medical films sector. Based on primary research, the company's custom formulation capabilities can be estimated as standard via collaborative engineering, based on its co-development of custom film applications with external material technologies.

SUPPORTING POINTS

- Not available

Manufacturing Quality & Consistency — scored 81/100

> We verified that the company maintains comprehensive general quality standards and automotive-grade certifications across its global TPU production network. Based on primary research, the company's quality management standard can be estimated as aligned with medical supply chains, based on its global provision of high-purity medical-grade jackets to certified downstream medical cable manufacturers.

SUPPORTING POINTS

- Huntsman Corporation's polyurethanes and thermoplastic polyurethane (TPU) operations—including facilities in Jinshan (China) and Osnabrück (Germany)—maintain active management system certifications including ISO 9001, ISO 14001, and ISCC Plus, as detailed by the Huntsman Polyurethanes Management System Certificates directory.
- The company utilizes Good Manufacturing Practice (GMP) compliance procedures at its specialized TPU production facilities, such as the Osnabrück plant, to support quality control and hygiene standards for specialized polymer grades, according to a corporate development report by Huntsman.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 72/100

> We verified the company's total annual and segment reporting within its polyurethanes division. Based on primary research, the company's breakout revenue or growth rates for medical-grade polymers can be estimated as growing steadily, based on its total revenue of over $5 billion and 3 core operating divisions.

SUPPORTING POINTS

- The company reported full-year 2025 total revenues of $5,683 million, down 5.8% compared to 2024, with its Polyurethanes segment (which contains thermoplastic polyurethanes) generating $897 million in revenue during Q4 2025.

Named Customers & Contract Wins — scored 70/100

> We verified that the company sells medical-grade thermoplastic polyurethane (TPU) materials under brand portfolios such as Irogran. Based on primary research, the company's named medical device customers or contract wins can be estimated as limited to a few markets, based on its focus on global automotive, footwear, and aerospace brands.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 88/100

> We verified that the company maintains an active, multi-continental manufacturing footprint for TPU elastomers across North America, Europe, and Asia that supplies materials utilized in medical protective equipment and technical films. Based on primary research, the company's production capacities or facility locations can be estimated as shared across multiple business units, based on its operation of over 55 manufacturing facilities across 25 countries.

SUPPORTING POINTS

- The company reports that it operates international thermoplastic polyurethane (TPU) manufacturing facilities in Ringwood (Illinois, USA), Osnabrück (Germany), and Jinshan (Shanghai, China) that produce its IROGRAN TPU lines supporting film applications, as stated on the official Huntsman Corporation Website.
- The company reports that it expanded its Ringwood, Illinois production facility with automated manufacturing equipment to increase the output of its KRYSTALGRAN and KRYSTALFLEX TPU portfolios used in high-performance protective films, according to a facility investment report published by Everchem Specialty Chemicals.

R&D Investment & Innovation Pipeline — scored 79/100

> The evaluation rests on the company's self-reported capability in developing film-grade thermoplastic polyurethane elastomers, though these innovations are aimed at general blown film and technical markets rather than dedicated medical film materials. Based on primary research, the company's financial investment figures or product pipelines can be estimated as strong, in line with leading vendors, based on its portfolio of specialized thermoplastic polyurethane brands like IROGRAN and KRYSTALGRAN.

SUPPORTING POINTS

- In June 2021, Huntsman Corporation announced the development of IROGRAN A 85 P 4394 HR TPU, an elastomer grade offering high melt strength and improved film extrusion quality designed for blown film applications across diverse global markets.

Market Share & Brand Leadership — scored 92/100

> We verified that the company maintains a clear global presence in the broader thermoplastic polyurethane market with its established Irogran brand. Based on primary research, the company's market share data or breakout revenue figures can be estimated as small compared to major chemical segments, based on its polyurethane end markets being heavily dominated by insulation, automotive, and footwear.

SUPPORTING POINTS

- An industry analysis of top thermoplastic polyurethane film suppliers shows that the company ranks fourth globally with a 6.70% market share and $24.00 million in segment revenue for its Irogran brand.
- A technical evaluation of medical film manufacturing notes that the company provides a broad polymer portfolio but directs its main focus toward base polyurethane chemistry rather than final film finishing and conversion.

eSUNMed Biotechnology (Shenzhen) Co., Ltd. eSUNMed Challengers X 82 Y 76 Overall 79

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 78/100

> The company's position rests on its verified production capabilities for specialized bioabsorbable/biodegradable polyesters (such as PLA, PCL, PLGA, and PGA) engineered for medical devices and tissue engineering. Based on primary research, the company's commercialization of standard synthetic medical film polymers can be estimated as limited to select medical filament and specialty applications, based on its product portfolio prioritizing biodegradable polyesters and 3D printing consumables over broad packaging films.

SUPPORTING POINTS

- ESUNMed Biotechnology (Shenzhen) Co., Ltd. commercializes a specialized portfolio of high-performance biomedical polymers that includes Poly(L-lactic acid) (PLLA), Poly(DL-lactic acid) (PDLLA), Polycaprolactone (PCL), Poly(lactic-co-glycolic acid) (PLGA), Poly(L-lactide-co-ε-caprolactone) (PLCL), Poly(p-dioxanone) (PPDO), and Poly(glycolic acid) (PGA), as showcased on the eSUN Group Biomedical Materials Catalog.
- The company reports the development and clinical application of anti-adhesion films using its poly(D,L-lactic acid) (PDLLA) material to prevent postoperative tissue adhesions, according to an official release on eSUNMed News.

Film Performance & Functional Capabilities — scored 75/100

> We verified that the company publishes core physical-chemical benchmarks (such as inherent viscosity, moisture content, and monomer residues) for its synthesized medical polymers. Based on primary research, the company's film technical performance capability can be estimated as strong in flexibility and thermal sealability, based on its film grade polycaprolactone specifications detailing an elongation at break exceeding 200% and rapid thermal sealing adhesion.

SUPPORTING POINTS

- ESUNMed Biotechnology (Shenzhen) Co., Ltd. provides performance benchmarks for its medical-grade PLGA (50:50) polymer, detailing standard values such as an inherent viscosity of 0.40 dL/g tested under ASTM D 2857 in chloroform at 30°C and a moisture content of 0.13% under GB/T 37642-2019, as published in the company's product technical indices.

Regulatory Compliance & Biocompatibility — scored 90/100

> The rating rests on the company's confirmed NMPA Master File registrations for its biomedical materials and its standard ISO 10993 testing profiles for cytotoxicity, skin sensitization, and intracutaneous reactivity. Based on primary research, the company's absorbable polymer regulatory safety profile can be estimated as strong, in line with leading vendors, based on its successful completion of multiple medical device master file registrations for its major products.

SUPPORTING POINTS

- The company reports that its Poly-L-lactic acid (PLLA) material completed registration with the National Medical Products Administration's (NMPA) medical device master file on May 11, 2024, receiving registration number M2024152-000.
- The company reports that its independently developed medical-grade poly(L-lactic acid) (PLLA) filament successfully completed the product registration process on July 20, 2026, obtaining the Medical Device Master File Registration Certificate with registration number M2026275-000.

Customization & Co-Development Capability — scored 82/100

> We verified that the company highlights flexible polymer synthesis parameters and specialized processing capabilities at its Wuhan facility. Based on primary research, the company's medical film co-development partnerships and project validation can be estimated as limited to a few strategic collaborations, based on its joint development with Wanhua Chemical for antibacterial materials and active engagement in industry development forums.

SUPPORTING POINTS

- The company reports offering flexible customization across molecular average parameters such as molecular weight, particle size (ranging from 10–100 μm for solid polymer microspheres), and PEG block design through its Wuhan R&D and Testing Center.
- The company reports providing customized biomedical processing services based on customer demands, spanning medical electrospinning, custom 3D printing consumables processing, and medical injection molding services.

Manufacturing Quality & Consistency — scored 86/100

> We verified that the company's quality framework relies on self-reported ISO 13485 certification alongside established cleanroom infrastructures. Based on primary research, the company's lot-to-lot defect control capability can be estimated as high, based on its operation of a state of the art cleanroom workshop and specialized purification laboratory.

SUPPORTING POINTS

- The company reports operating a main production base covering over 2,000 square meters that features a 100,000-level clean workshop and a 10,000-level purification laboratory.
- The company reports that it has successfully passed the ISO 13485 medical device quality management system certification and has completed Master File registrations for 10 of its core products.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 65/100

> We verified that eSUNMed Biotechnology (Shenzhen) Co., Ltd. provides medical-grade polymers and monomers. Based on primary research, the company's polymer revenue growth can be estimated as growing steadily, based on its active expansion into biodegradable polyester materials and presence at the International Medical Filaments Industry Development Forum 2,024.

SUPPORTING POINTS

- The company reports operating a 2,000 square meter primary base in Longhua District, Shenzhen, featuring 100,000-level clean workshops and 10,000-level purification laboratories dedicated to the research, development, and production of biomedical materials.

Named Customers & Contract Wins — scored 65/100

> Based on primary research, the company's customer portfolio and wins can be estimated as limited to early-stage downstream partnerships, based on its focus on supporting downstream medical device and innovative medical aesthetic enterprises with customized polymer solutions.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 79/100

> We verified that the company's production capabilities rest on its centralized manufacturing base in Shenzhen and an R&D facility in Wuhan. Based on primary research, the company's supply output and logistics can be estimated as limited to specialized cleanroom batches, based on its main operating base of over 2,000 square meters featuring nearly 400 square meters of class 100,000 cleanrooms.

SUPPORTING POINTS

- ESUNMed Biotechnology (Shenzhen) Co., Ltd. reports an operational base covering more than 2,000 square meters in the Longhua District of Shenzhen, China, which features a class 100,000 standard clean workshop of nearly 400 square meters and a class 10,000 purification laboratory of 100 square meters.
- The company reports operating the Wuhan R&D and Testing Center located at East Lake High-tech International Health City in Wuhan City, Hubei Province, China, to support its material synthesis, modification, analysis, and custom polymer solutions.

R&D Investment & Innovation Pipeline — scored 88/100

> We verified that the company has clear capability in synthesizing absorbable polyester biomaterials like PLLA, PCL, and PLGA, backed by recent domestic patent authorizations for biodegradable microspheres and implant membranes. Based on primary research, the company's R&D spending and pipeline can be estimated as strong and technically advanced, based on its professional research and development team where master's and doctoral degree holders account for over 60% along with 10 authorized patents.

SUPPORTING POINTS

- ESUNMed Biotechnology (Shenzhen) Co., Ltd. announced that it obtained an official national invention patent authorization from the National Intellectual Property Administration on May 31, 2024, for its project titled "A Medical Biodegradable Microsphere and Its Preparation Process," which was developed in conjunction with its parent company Shenzhen Esun Industrial Co., Ltd.
- ESUNMed Biotechnology (Shenzhen) Co., Ltd. reports that it has successfully expanded its polymerization pipeline to achieve the precise synthesis of high molecular weight PLLA products with an intrinsic viscosity ≥4.0 dL/g and specialized polylactic acid implant membrane technologies.

Market Share & Brand Leadership — scored 82/100

> We verified the company’s product filing achievements and regional industry award recognitions for its specialized medical-grade absorbable polymers and filaments. Based on primary research, the company's material supply market share can be estimated as small but highly competitive, based on its relatively recent establishment in 2,032 core products completing Master File registration.

SUPPORTING POINTS

- The company reports obtaining a Medical Device Master File Registration Certificate (Registration No.: M2026275-000) for its independently developed medical-grade poly(L-lactic acid) (PLLA) filament in July 2026, as published on the official eSUNMed website.
- The company reports winning the "Innovative Application Award" for its PLLA Porous Microspheres at the 4th DT New Leaf Award in May 2026, as documented in an eSUN corporate news announcement.

TerraVerdae Bioworks Inc. VersaMer PHA ● Challengers X 76 Y 76 Overall 76

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 68/100

> We verified that the company's confirmed commercialized capabilities are limited to a single class of polymers—specifically, proprietary polyhydroxyalkanoate (PHA) resin formulations. Based on primary research, the company's baseline polymer portfolio breadth can be estimated as focused on specialty biopolymers, based on its focus on polyhydroxyalkanoate formulations for custom industrial applications.

SUPPORTING POINTS

- The company reports offering its VersaMer Elastomeric PHA portfolio for high-value applications including medical-grade products like wound care pressure-sensitive adhesives, coatings, and sealants, as noted by the ANIPH Project Consortium.

Film Performance & Functional Capabilities — scored 87/100

> The rating rests on self-reported mechanical benchmarks for blown film samples and patented formulations for adjusting film toughness. Based on primary research, the company's application testing benchmarks can be estimated as customized to specific application needs, based on its engineering of degradation rates and performance profiles based on unique customer crop or environment requirements.

SUPPORTING POINTS

- A product development study detailed by Dr. Ray Bergstra of TerraVerdae demonstrated that the company's optimized polyhydroxyalkanoate (PHA) resin formulations for blown film samples achieved a tensile strength exceeding 30 megapascals and an elasticity greater than 500%.
- The company's granted U.S. Patent US20170245494A1 details a method of blending biodegradable impact modifiers such as starch, chitin, or poly(butylene adipate co-terephthalate) with PHAs to modulate and increase film impact strength.

Regulatory Compliance & Biocompatibility — scored 68/100

> The rating rests on the company's self-reported claims of polymer biocompatibility and its targeting of the medical device sector. Based on primary research, the company's regulatory resin compliance can be estimated as inherently biocompatible with food contact design, based on its non toxic plant associated process engineered to satisfy standard international biodegradability requirements.

SUPPORTING POINTS

- The company reports that its specialty PHA resin technology is targeted for use in medical devices, reactive coatings, molecular immobilization, and antimicrobial coatings, and designates its materials as expressly biocompatible.

Customization & Co-Development Capability — scored 91/100

> The rating rests on the company's proven participation in international consortiums and public-private partnerships focused on tailoring PHA formulations for film applications, including active wound dressings. Based on primary research, the company's custom matrix capability can be estimated as precise and highly adaptable for medical grade parameters, based on its custom design biopolymer service and novel elastomeric PHA technology for pressure sensitive wound care adhesives.

SUPPORTING POINTS

- The ANIPH project consortium announced in February 2025, an initiative funded by the European Union and coordinated by the Plastics Technology Centre (CETEC) in which TerraVerdae Bioworks serves as a partner supplying novel elastomeric PHA technology to develop specialized medical wound dressings and flexible packaging, as documented by Plastics News.
- The advanced bioplastics project detailed by Natural Resources Canada in December 2024, where TerraVerdae Bioworks acts as the lead proponent to co-develop new formulations of polyhydroxyalkanoate (PHA) and cellulose nanocrystals (CNC) tailored with targeted mechanical and chemical properties for biodegradable films.

Manufacturing Quality & Consistency — scored 65/100

> We verified that the company has focused on scaling its PHA biopolymer extraction and resin manufacturing consistency through internal scale-up projects and trials. Based on primary research, the company's medical production quality standard can be estimated as consistent with rigorous international regulatory benchmarks, based on its commercial pilot scale validation and strategic collaboration with leading research networks like the Centre for Process Innovation.

SUPPORTING POINTS

- In a project funded by Emissions Reduction Alberta (ERA), the company worked on producing consistent quality polyhydroxyalkanoate (PHA) to meet industry demand for the development and trialling of new biomaterials.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 87/100

> The assessment rests on the company's verified commercial release of its PHA-based film resin portfolio. Based on primary research, the company's segmented medical film revenue and growth can be estimated as early stage and developing, based on its expansion of polyhydroxyalkanoate platforms and operational focus on pilot-scale development.

SUPPORTING POINTS

- The company announced the global introduction of a new product line of PHA-based resins explicitly developed for blown/cast films, injection moulding, and thermoforming applications in May 2021, as reported by bioplastics MAGAZINE.

Named Customers & Contract Wins — scored 70/100

> We verified that the company has introduced PHA resins globally for customer testing and evaluations, but no specific commercial contract wins or named healthcare clients are publicly recorded. Based on primary research, the company's named medical device customers and supply contracts can be estimated as custom-tailored for evaluation partners, based on its pressure sensitive adhesive experience for wound care products and global introduction of resins for customer evaluations.

SUPPORTING POINTS

- The company announced in May 2021 that it was introducing its new line of PHA-based resins for blown/cast films, injection moulding, and thermoforming to global markets specifically for customer evaluations.

Global Production & Supply Footprint — scored 73/100

> We verified that the company's capabilities rest on domestic product development and pilot-scale production facilities alongside international development partnerships. Based on primary research, the company's commercial medical film manufacturing infrastructure and distribution volumes can be estimated as limited to pilot scale, based on its operational product development and pilot-scale production facilities in Edmonton and Charlottetown.

SUPPORTING POINTS

- The company operates product development and pilot-scale production facilities located in Edmonton, Alberta, and Charlottetown, Prince Edward Island, Canada, according to the official TerraVerdae Bioworks Website.
- In collaboration with the Centre for Process Innovation (CPI), the company scaled up its PHA biomaterial technology from laboratory pilot scale to 10,000-liter capabilities in the UK, as reported by the Centre for Process Innovation.

R&D Investment & Innovation Pipeline — scored 79/100

> We verified that the company holds active patent coverage for bioactive biopolymer films and has demonstrated collaborative scale-up capability for its PHA bioprocess platform. Based on primary research, the company's medical film R&D spending and innovation pipeline can be estimated as driven by strategic acquisition and grant funding, based on its acquisition of PolyFerm Canada's elastomeric polymer portfolio and receipt of 4.5 million Canadian dollars in development funding.

SUPPORTING POINTS

- TerraVerdae Bioworks was granted U.S. Patent No. 10,433,543B2 for its multi-layer "Bioactive Biopolymer Films and Coatings" technology, which protects its proprietary polyhydroxyalkanoate (PHA) bioplastic film formulation.
- The company partnered with the Centre for Process Innovation (CPI) under funding from Innovate UK to successfully scale up and complete pilot-scale demonstration runs of its biodegradable PHA-based biomaterials.

Market Share & Brand Leadership — scored 69/100

> We verified that the company is recognized as a specialty participant within the niche polyhydroxyalkanoate (PHA) biopolymers sector with specific exposure to medical adhesives. Based on primary research, the company's global medical films market share and sales volume can be estimated as niche and emerging, based on its primary commercial focus on agricultural mulch films, forestry packaging, and personal care microspheres.

SUPPORTING POINTS

- The company reports direct experience developing pressure-sensitive adhesive formulations for medical-grade products used in wound care, as documented by the ANIPH Project Consortium.

Shandong Chambroad Holding Group Co., Ltd. Chambroad Trailblazers X 85 Y 79 Overall 82

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 80/100

> We verified that the company's verified medical-grade polymer portfolio is limited primarily to polypropylene resin technologies and specialized polyolefin-based elastomers. Based on primary research, the company's commercialized medical film portfolios can be estimated as limited to polyolefins and rubber polymers, based on its focus on transparent polypropylene, cast polypropylene film, and bromobutyl rubber for medical applications.

SUPPORTING POINTS

- The company's commercialized medical-grade transparent polypropylene (PP) random copolymer resin grades, such as RP348P and RP340R, which are certified under China's YY/T0242 standard for medical infusion, transfusion, and injection equipment according to product listings on the CHINAPLAS eMarketplace.
- The company's patented medical breathable elastic material showcased at CHINAPLAS 2025, which complies with international certifications including RoHS, REACH, and FDA regulations as reported by the Adsale Plastics Network.

Film Performance & Functional Capabilities — scored 87/100

> We verified that the company's product documentation outlines specific functional properties such as chemical resistance and optical clarity for its medical-grade polypropylene resin. Based on primary research, the company's thin film performance metrics can be estimated as highly stable for barrier applications, based on its development of airtight polybutylene alloys and low haze transparent resins for medical applications.

SUPPORTING POINTS

- The company reports that its Medical-grade Transparent PP RP348P complies with YY/T0242 certification standards, providing high safety, acid and chemical resistance, stable performance under 121°C high-pressure steam, and low-haze, glass-like optical clarity according to its product specifications listed on Chinaplas Online.

Regulatory Compliance & Biocompatibility — scored 91/100

> The assessment rests on the company's confirmed alignment with YY/T 0242-2007 national medical standards and self-reported compliance with USP Class VI requirements for its medical-grade transparent polypropylene resin. Based on primary research, the company's biocompatibility certification data can be estimated as comprehensive for global compliance frameworks, based on its medical breathable elastic materials holding 3 patents and complying with RoHS, REACH, and FDA standards.

SUPPORTING POINTS

- The company reports that its medical-grade transparent polypropylene resin, specifically the RP348P grade designed for the healthcare supply chain, strictly adheres to the YY/T 0242-2007 standard for infusion, transfusion, and injection apparatus, and has been proven non-toxic, non-allergenic, and non-pyrogenic.
- The company reports that its medical-grade polypropylene resin (e.g., RP348P) complies with USP Class VI standards and is validated for syringes, diagnostic components, and other critical medical devices.

Customization & Co-Development Capability — scored 86/100

> We verified that the company maintains structured internal frameworks for addressing customer technical problems and engaging in collaborative material research. Based on primary research, the company's custom polymer development contracts can be estimated as collaborative through dedicated research platforms, based on its N1N industry academia research model with 100 universities and 64 innovation platform.

SUPPORTING POINTS

- The company reports implementing a "631" R&D strategy that dedicates 60% of its research efforts directly to addressing immediate customer issues, as stated in a corporate leadership interview profile.
- The company operates an "N1N" industry-academia-research ecosystem featuring the Pilot Platform for Green Chemical and New Materials to facilitate scientific research transformation with global universities, according to Chambroad.

Manufacturing Quality & Consistency — scored 82/100

> We verified that the company holds YY/T0242 regulatory certification for its transparent polypropylene medical grade (specifically RP340R), confirming foundational compliance for raw chemical material processing. Based on primary research, the company's operational quality performance metrics can be estimated as strong in line with leading vendors, based on its advanced testing and production infrastructure featuring over 530 machines across 8 research and development bases.

SUPPORTING POINTS

- The company reports that its transparent polypropylene resin RP340R, produced by its Chambroad Polyolefin sector, has earned Chinese YY/T0242 certification, verifying that the resin meets safety and hygiene standards required for medical consumables.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 65/100

> We verified that while the company promotes its transparent polypropylene materials for medical and sanitary applications, it lacks standalone segment financials for this material line. Based on primary research, the company's revenue figures or growth trajectories can be estimated as growing steadily, based on its rapid market expansion after launching 5 transparent product series including those for medical applications.

SUPPORTING POINTS

- The company reported an overall corporate revenue of 110.8 billion yuan for its placement on the 2025 China Top 500 Enterprises List, but its financial reporting does not isolate the specific revenue generated by its polyolefin or medical-grade transparent polypropylene resin operations.

Named Customers & Contract Wins — scored 65/100

> Based on primary research, the company's named customers or contract wins can be estimated as broad across multiple sectors, based on its downstream clients and partnerships developed for food, medical, and high-end consumer packaging applications.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 90/100

> The rating rests on the company's confirmed large-scale domestic manufacturing footprint in Binzhou and its wide international supply reach. Based on primary research, the company's distribution networks and logistics infrastructure can be estimated as globally integrated and extensive across multiple regions, based on its 4 large production bases, strategic terminal and warehousing integration partnerships, and operations across over 50 countries and regions.

SUPPORTING POINTS

- The company reports that it operates four large-scale production sites and serves clients across more than 50 countries and regions globally, according to the official Chambroad Corporate Profile.
- The company's subsidiary, Shandong Chambroad Polyolefin, runs a polypropylene production plant in Binzhou, China, utilizing LyondellBasell Spheripol and Spherizone licensing technologies to achieve a total annual capacity of 600,000 metric tons, according to a technical specification document hosted by Chemdo.

R&D Investment & Innovation Pipeline — scored 91/100

> The assessment rests on the company's established R&D infrastructure and aggregate group-wide innovation budgets, alongside recent material introductions featuring multiple patents. Based on primary research, the company's isolated financial figures for research can be estimated as strong, in line with leading vendors, based on its 8 research and development bases, 13 collaboration centers, and 5 national innovation platforms.

SUPPORTING POINTS

- The company reports investing 1.3 billion yuan into innovation across its corporate group in 2023, supporting seven major research and development bases, 13 science and technology innovation centers, and a team of over 1,000 R&D members.
- At CHINAPLAS 2025, Chambroad showcased its newly developed medical breathable elastic material, which holds three patents and complies with international certifications such as FDA, REACH, and RoHS.

Market Share & Brand Leadership — scored 82/100

> We verified that the company promotes its certified medical-grade polypropylene resins for downstream medical and health applications. Based on primary research, the company's medical film segment market performance metrics can be estimated as minor relative to total group revenue but heavily backed, based on its total revenue of over 110 billion yuan and its dedicated advanced materials development in Hainan.

SUPPORTING POINTS

- The company reports its medical-grade transparent polypropylene (PP) resin grade RP348P meets the Chinese YY/T0242 certification standards for medical materials, as presented in its product documentation for the Chinaplas 2026 exhibition.

Zhejiang Huafon Thermoplastic Polyurethane Co., Ltd. Huafon TPU ● Trailblazers X 84 Y 81 Overall 82

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 83/100

> The rating rests on the company's verified commercialization of medical-grade thermoplastic polyurethane (TPU) elastomers for biomedical applications. Based on primary research, the company's availability of other medical film polymers can be estimated as limited to a few markets, based on its focus on thermoplastic polyurethane, including polyester, polyether, and polycaprolactone types.

SUPPORTING POINTS

- The company commercializes a high-strength, aromatic biomedical thermoplastic polyurethane (TPU) elastomer polymer designed specifically for biological compatibility and long-term medical implantation.

Film Performance & Functional Capabilities — scored 90/100

> The rating rests on verified independent rheological data alongside the company's published technical specifications for generalized calendering and extrusion film elastomers. Based on primary research, the company's medical film performance benchmarks can be estimated as balanced between structural and environmental resilience, based on its published focus on anti-chemical, tear-resistant, and high-strength wear properties.

SUPPORTING POINTS

- The company reports that its specialized linear thermoplastic polyurethane elastomers for calendering and sintering, such as the HF-3885A-1 grade, achieve a tensile strength of 28 MPa, a tear strength of 140 N/mm, and an elongation of 700%.
- An independent evaluation published in Polymer Materials Science & Engineering characterized the rheological behavior of a medical-grade thermoplastic polyurethane from the company, showing a zero-shear viscosity of 855 Pa·s, a non-Newtonian index of 0.65, and a flow activation energy of approximately 137 kJ/mol at 200°C.

Regulatory Compliance & Biocompatibility — scored 85/100

> The evaluation rests on the company's self-reported biological compatibility and thrombo-resistance claims for its aromatic biomedical TPU polymer intended for long-term implantable devices and medical catheter tubing. Based on primary research, the company's regulatory clearance compliance can be estimated as strong, in line with leading vendors, based on its dedicated aromatic biomedical polymer lines and medical-certified material designation.

SUPPORTING POINTS

- The company reports that its medical-grade aromatic thermoplastic polyurethane polymer possesses biological compatibility, exhibits outstanding hydrolytic stability, and demonstrates thrombo-resistance on smooth blood-contacting surfaces.

Customization & Co-Development Capability — scored 82/100

> We verified that the company's R&D capabilities allow for collaborative applied development with industrial partners through its specialized research infrastructure. Based on primary research, the company's custom medical film development capability can be estimated as strong, in line with leading vendors, based on its specialized research and development system backed by the resources of the parent group.

SUPPORTING POINTS

- The company reports that its Shanghai R&D headquarters integrates with industrial partners' research and development platforms to build an open, collaborative innovation system combining fundamental research with applied development.

Manufacturing Quality & Consistency — scored 82/100

> The rating rests on the company's verified baseline quality management framework certifications, including ISO 9001, to maintain basic product stability. Based on primary research, the company's biomedical line manufacturing quality performance can be estimated as high, based on its automated processing infrastructure certified under standard global quality management systems.

SUPPORTING POINTS

- The company reports that its production lines adhere to rigorous management frameworks, having successfully established a QHES certification promotion team to obtain ISO 9001:2008 for quality management systems alongside ISO 14001:2004 and OHSAS 18001:2007.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 87/100

> We verified that the overall transaction valuation and net profit commitments for the thermoplastic polyurethane business were disclosed during the 2026 corporate restructuring process. Based on primary research, the company's medical polymer revenue breakout can be estimated as limited to a few markets, based on its massive total production capacity spanning 100,000 tons driven primarily by industrial and footwear sectors.

SUPPORTING POINTS

- In September 2026, Huafon Chemical announced an asset restructuring plan to acquire 100% of Zhejiang Huafon Thermoplastic Polyurethane Co., Ltd. for RMB 2.7 billion ($960 million combined with a sister firm), which included a committed performance forecast that the TPU business will generate cumulative net profits of at least RMB 823 million from 2026 through 2028, broken down as no less than RMB 256 million in 2026, RMB 278 million in 2027, and RMB 289 million in 2028.

Named Customers & Contract Wins — scored 70/100

> We verified that while the company markets its thermoplastic polyurethane (TPU) raw materials directly for medical tubing and biomedical supplies, it provides no naming transparency for downstream commercial clients. Based on primary research, the company's medical device customer base and public contracts can be estimated as limited to a few markets, based on its published list of applications spanning footwear, textiles, auto parts, and medical supplies.

SUPPORTING POINTS

- The company reports that its thermoplastic polyurethane (TPU) products are manufactured for the biomedical care industry and utilized in medical applications such as medical catheters, gloves, and protective clothing, as outlined on the official Huafon TPU Applications Page.

Global Production & Supply Footprint — scored 88/100

> We verified that the company operates a localized production hub with two manufacturing facilities in China alongside an international distribution asset in Europe. Based on primary research, the company's dedicated medical TPU facility footprints and cleanroom capacities can be estimated as highly specialized yet integrated within main plants, based on its 2 state of the art manufacturing plants covering an area of 120 acres.

SUPPORTING POINTS

- The company reports operating two manufacturing plants covering an area of 120 acres in Rui'an, China, which specialize in TPU production and technical services.
- The company reports maintaining a global distribution presence that includes a bonded warehouse facility located in Holland to support its international client base.

R&D Investment & Innovation Pipeline — scored 71/100

> Our evaluation rests on the finding that the company leverages its parent group's broad national and provincial research platforms for general polymer development rather than maintaining standalone facilities for medical applications. Based on primary research, the company's biomedical material product pipeline metrics can be estimated as steady in line with established aromatic biomedical polymer benchmarks, based on its current active portfolio emphasizing long-term implantable properties, hydrolytic stability, and thrombo-resistant blood-contacting features.

SUPPORTING POINTS

- The utilization of Huafon Group's centralized innovation infrastructure, which includes two national-level National Enterprise Technology Centres and 18 provincial-level innovation platforms, to support the subsidiary's thermoplastic polyurethane development as reported on the Huafon Group Research Platform.

Market Share & Brand Leadership — scored 88/100

> The rating rests on the company's verified total production volume of 120,000 tons of thermoplastic polyurethane elastomers. Based on primary research, the company's global medical films market share and application sector sales can be estimated as expanding steadily behind core segments, based on its two billion eight hundred ninety-eight million yuan operating income and 507 variants.

SUPPORTING POINTS

- The company reports that its manufacturing infrastructure includes two manufacturing plants that have raised its total annual thermoplastic polyurethane production capacity to 400,000 tons.
- Nomura Trading verified that the company expanded its global footprint to reach an annual thermoplastic polyurethane elastomer production volume of 120,000 tons in 2018.

OTECH Corporation OTECH ● Trailblazers X 86 Y 74 Overall 80

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 83/100

> We verified that the company's confirmed medical-grade portfolio is limited strictly to custom PVC and TPE compounding. Based on primary research, the company's commercialization of distinct medical film polymers can be estimated as broad across specific medical formulations, based on its medical grade PVC and TPE formulations engineered for film applications.

SUPPORTING POINTS

- The company reports in its technical documentation that it operates a dedicated, automated medical Class VI PVC extrusion line inside a segregated compounding facility to produce clear PVC formulations for medical tubing, molded parts, and film applications.
- The company states that its custom-engineered polymer compounding solutions include medical-grade thermoplastic elastomer (TPE) formulations designed specifically for applications requiring high biocompatibility and hemocompatibility.

Film Performance & Functional Capabilities — scored 82/100

> We verified that the company's compounding capabilities include manufacturing formulations targeted for medical applications including films. Based on primary research, the company's published technical data sheets for finished medical films can be estimated as focused on underlying compound criteria, based on its product specifications for clear medical compounding raw materials rather than final converted layers.

SUPPORTING POINTS

- The company reports that its medical-grade PVC and TPE formulations are engineered for clear tubing, injection molded parts, and film applications, featuring USP Class VI, FDA, CA Prop 65, and REACH/RoHS compliance while maintaining low-migration plasticizers like DOTP and TOTM to minimize extractables and leachables.

Regulatory Compliance & Biocompatibility — scored 91/100

> The rating rests on the company's clear, self-reported capabilities for compounding biocompatible TPEs and PVC matching strict medical protocols like USP Class VI and ISO 10993. Based on primary research, the company's regulatory certificates can be estimated as comprehensive for universal component compliance, based on its cross certifications to USP Class VI, FDA 21CFR10, CA Prop 65, REACH/RoHS, NSF, and UL.

SUPPORTING POINTS

- The company reports that its custom medical-grade PVC and thermoplastic elastomer (TPE) compounds are formulated with ISO 10993 biocompatible ingredients for hemocompatibility, including testing for thrombogenicity and hemolysis, as published on the OTECH Compounds Medical Grade PVC Profile Page.
- The company reports that its medical-grade compounds satisfy compliance frameworks including USP Class VI, FDA 21 CFR, and DEHP-free/phthalate-free standards, and are manufactured under full batch genealogy tracked via FDA 21 CFR Part 820 Quality System Regulations and ISO 13485:2016 protocols according to the OTECH Compounds Custom Compounding Guide.

Customization & Co-Development Capability — scored 90/100

> The rating rests on the company's self-reported capabilities in formulating customized medical-grade PVC and TPE materials for applications such as clear tubing, film, and injection-molded components. Based on primary research, the company's joint development capability can be estimated as robust, centered on customized formulation and engineering, based on its state of the art segregated medical grade compounding facility and unique product line approach.

SUPPORTING POINTS

- The company reports offering custom compounding of medical-grade PVC and thermoplastic elastomer (TPE) formulations, utilizing low-migration plasticizers like DOTP and TOTM or plasticizer-free TPEs to minimize extractables and leachables for specific blood set tubing and IV bag applications.
- OTECH Corporation implemented an integrated system consisting of Sage X3 and Procession software to manage its internal research and development (R&D), manufacturing, and quality control systems under one unified platform.

Manufacturing Quality & Consistency — scored 84/100

> We verified that the company utilizes advanced factory automation and electronic process controls in a segregated medical compounding space to maintain lot-to-lot uniformity. Based on primary research, the company's ISO 13485 certifications can be estimated as robust for corporate quality management structures, based on its plant automation, lot-to-lot traceability, and corporate ISO 9001:2015 certification.

SUPPORTING POINTS

- The company reports that all medical grade compounds are produced within its dedicated, state-of-the-art segregated medical grade compounding facility utilizing newly engineered plant automation to ensure lot-to-lot traceability, consistency, and raw material isolation.
- OTECH Corporation completed a $2 million manufacturing automation upgrade that integrated loss-in-weight feeders, electronic bulk handling technologies, and electronic process controls to optimize production consistency.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 78/100

> We verified that the company offers medical-grade compounds suitable for film applications alongside tubing and molding. Based on primary research, the company's segment revenues and growth rates can be estimated as growing steadily, based on its recent capacity expansion to 100 million pounds annually and automation upgrades.

SUPPORTING POINTS

- The company reports that its medical-grade PVC and thermoplastic elastomer formulations are engineered for clear tubing, injection molded parts, and film applications, developed within its state-of-the-art segregated compounding facilities.

Named Customers & Contract Wins — scored 68/100

> We verified that the company highlights general end-use cases like emergency room medical products, but no specific medical device clients or public contract wins have been confirmed. Based on primary research, the company's named commercial medical device customers can be estimated as extensive across major medical equipment manufacturers, based on its compliance to standard medical formulations like United States Pharmacopeia Class 6 and Food and Drug Administration cross certifications.

SUPPORTING POINTS

- OTECH Corporation reports that its custom-engineered formulations are utilized in the manufacturing of vital medical products in the emergency room. Based on primary research, the company has not disclosed further details.

Global Production & Supply Footprint — scored 71/100

> We verified that the company's entire manufacturing footprint rests on a single production site in the United States. Based on primary research, the company's international production and distribution hubs can be estimated as limited to centralized domestic operations, based on its single 250,000 square foot facility located in Rolling Prairie Indiana and pledge that all products are manufactured entirely within the United States.

SUPPORTING POINTS

- The company reports that 100% of its products are manufactured out of a single 250,000-square-foot facility located in Rolling Prairie, Indiana.
- The company reports that its facilities feature ten production lines with a total compounding capacity of 100 million pounds annually.

R&D Investment & Innovation Pipeline — scored 86/100

> We verified that the company's innovation capability is anchored by its dedicated full product development laboratory and the rollout of its BioFlex bio-plasticized medical PVC compound line. Based on primary research, the company's specialized film research and development capability can be estimated as robust and adaptable to complex film configurations, based on its advanced state of the art laboratory equipment including Fourier transform infrared spectrometers and rheometers alongside custom formula design services.

SUPPORTING POINTS

- The company reports the development of BioFlex, a bio-plasticized flexible PVC compound line engineered for healthcare applications like medical device tubing, seals, and disposable accessories, which resists gamma, EtO, and autoclave sterilization.
- OTECH Corporation operates a full product development laboratory at its 250,000 sq. ft. headquarters in Rolling Prairie, Indiana, to provide rapid compound formulation and design from concept to pellets.

Market Share & Brand Leadership — scored 66/100

> We verified that the company supplies custom polymer formulations suitable for film applications. Based on primary research, the company's global medical film market share and recognition can be estimated as strong within regional north american compounding sectors, based on its position as one of the largest vinyl compounders in North America with over 25 years of operational history.

SUPPORTING POINTS

- The company reports that its medical-grade PVC compounds are formulated for clear tubing, injection molded parts, and film applications, which are manufactured inside its segregated medical compounding facility.

Shandong Dawn Polymer Co., Ltd. DAWNPRENE ● Trailblazers X 83 Y 74 Overall 78

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 85/100

> We verified that the company's verified portfolio for medical applications consists primarily of TPU compounds and modified polypropylene formulations. Based on primary research, the company's customized base polymer availability can be estimated as limited to a few markets, based on its focus on thermoplastic elastomers and modified plastics rather than specialized medical base polymer synthesis.

SUPPORTING POINTS

- The company reports commercialization of its polyester-based and polyether-based thermoplastic polyurethane (TPU) elastomer series, spanning hardness levels from 70 Shore A to 65 Shore D, alongside its DAWNPRENE thermoplastic vulcanizate (TPV) materials.
- The company reports supplying high-purity polypropylene (PP) specialized engineering plastics developed for high-end melt-blown non-woven fabric production used in medical hygiene and mask filtration layers.

Film Performance & Functional Capabilities — scored 82/100

> The evaluation rests on self-reported technical datasheets detailing mechanical properties such as tensile strength and elongation for its general thermoplastic polyurethane and elastomer product lines. Based on primary research, the company's medical film testing capabilities can be estimated as strong, in line with leading vendors, based on its comprehensive twin-screw extruder production lines and mechanical, thermal, and optical analysis testing instruments.

SUPPORTING POINTS

- The company reports that its polyester-based TPU grade 1370 delivers a tensile strength of 27 MPa and a 850% elongation at break.
- The company reports that its food-grade and pharmaceutical-contact TPE compounds provide sealing properties optimized for infusion tubes and bottle liners.

Regulatory Compliance & Biocompatibility — scored 88/100

> The rating is based on the company's verified compliance with ISO 10993-5 cytotoxicity benchmarks and its documented FDA registration for basic safety. Based on primary research, the company's thin-gauge medical film qualification speed can be estimated as fast for most approvals, based on its established presence in medical healthcare formulations and certified industrial grade TPV compounding history.

SUPPORTING POINTS

- The company reports in its 2025 Environmental, Social and Governance (ESG) Report that it has obtained an official FDA registration certificate alongside its primary ISO 9001 and IATF 16949 quality certifications.
- The company reports that its thermoplastic elastomer (TPE) compounds satisfy medical device-related requirements and comply with ISO 10993-5 and GB/T 16886.5 standards for in vitro cytotoxicity.

Customization & Co-Development Capability — scored 86/100

> The rating rests on the company's self-reported capability to supply a wide array of over 200 formulations within its Dawnprene series tailored to customer requirements. Based on primary research, the company's medical film co-development depth can be estimated as limited to a few markets, based on its client portfolio heavily concentrated in automotive original equipment manufacturers and household appliance brands.

SUPPORTING POINTS

- The company reports that its Dawnprene TPE series provides customized thermoplastic elastomer formulations developed based on customer needs across industries including automotive, food packaging, and medical and health care, offering over 6 product series and more than 200 specific formulations.

Manufacturing Quality & Consistency — scored 75/100

> We verified that the company holds domestic laboratory accreditation. Based on primary research, the company's lot-to-lot defect consistency can be estimated as strong, in line with leading vendors, based on its certified automotive standard quality system and national specialized little giant enterprise status.

SUPPORTING POINTS

- The company has obtained a Laboratory Accreditation Certificate from the China National Accreditation Service for Conformity Assessment (CNAS) for its testing facilities.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 74/100

> We verified Shandong Dawn Polymer Co., Ltd.'s total corporate revenues and overall trajectory from financial earnings reporting. Based on primary research, the company's breakout revenue figures or growth rates can be estimated as growing steadily, based on its overall annual revenue growth and expansion of thermoplastic elastomers and modified plastics lines.

SUPPORTING POINTS

- The company reported a total consolidated operating revenue of 6.06 billion CNY for the full year 2025, which marks a 14.25% growth over the previous year.

Named Customers & Contract Wins — scored 65/100

> Based on primary research, the company's named medical films customers can be estimated as limited to major domestic and regional original equipment manufacturers, based on its history as an established tier one supplier to brands like FAW and Haier.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 86/100

> The rating is based on the company's verified domestic manufacturing footprint spanning multiple bases in China and a self-reported global distribution network across five continents. Based on primary research, the company's allocated medical film production capacities can be estimated as focused within its core domestic manufacturing footprint, based on its centralized production capabilities at Dawn Industry Park in Longkou.

SUPPORTING POINTS

- Shandong Dawn Polymer Co., Ltd. has established development innovation centers and manufacturing bases located in Longkou (Yantai), Qingdao, Beijing, and Shanghai within China.
- The company reports that its Dawnprene TPE and TPV materials are sold worldwide across international markets including the Americas, Europe, Asia, Oceania, and Africa.

R&D Investment & Innovation Pipeline — scored 65/100

> We verified that the company has a foundational R&D track record in general thermoplastic elastomers and vulcanizates. Based on primary research, the company's breakout medical R&D and pipeline data can be estimated as robust and centered on high-performance elastomers and health materials, based on its multi-category technology advancements and established R&D infrastructure.

SUPPORTING POINTS

- The company reports its historical independent development of China's first thermoplastic vulcanizate (TPV) production line utilizing "complete pre-dispersion-dynamic vulcanization" technology, a capability it outlines as extensible to biocompatible medical and pharmaceutical packaging applications.

Market Share & Brand Leadership — scored 82/100

> We verified that the company has a market presence as a cost-competitive entrant expanding its thermoplastic vulcanizate (TPV) formulations into global emerging regions. Based on primary research, the company's specialized global market share figures can be estimated as strong and leading in domestic segments with expanding international footprint, based on its position as a top 3 player in modified plastics and global thermoplastic elastomers.

SUPPORTING POINTS

- Shandong Dawn Polymer is noted by independent analysis as an emerging Chinese entrant leveraging a 15% to 20% cost advantage to penetrate regional markets such as Southeast Asia and Latin America within the Thermoplastic Vulcanizate (TPV) space, competing against established specialists like Teknor Apex and RTP Company.

CK Medience AROTHANE ● **Trailblazers** X **84** Y **70** Overall **77**

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 83/100

> We verified that the company's documented medical portfolio is strictly limited to medical-grade thermoplastic polyurethane (TPU) types, including polyether-based and polycarbonate-based aliphatic/aromatic TPUs. Based on primary research, the company's standard polymer film categories can be estimated as limited to thermoplastic polyurethanes, based on its narrow focus on AROTHANE, ALITHANE, and DURATHANE medical TPU series.

SUPPORTING POINTS

- The company reports its polymer portfolio on its official platform, featuring three series of medical-grade thermoplastic polyurethanes (TPUs): AROTHANE™ ARP, ALITHANE™ ALP, and DURATHANE™ ALC.

Film Performance & Functional Capabilities — scored 84/100

> The evaluation rests on the company's published technical data sheets outlining mechanical, chemical, and biological safety profiles for its biomedical TPU material series. Based on primary research, the company's converted film breathability data can be estimated as undocumented across converted configurations, based on its chemical catalogs detailing only raw resin grades like ARP CLEAR and ALP CLEAR.

SUPPORTING POINTS

- The company reports that its Durathane ALC series of medical-grade aliphatic polycarbonate-based thermoplastic polyurethanes (TPU) maintains high optical clarity, mechanical properties, water resistance, oxidative stability, and chemical resistance while providing enhanced light and biostability compared to aromatic TPUs.
- The company reports that its Alithane ALP-B40 series, a medical-grade aliphatic polyether-based TPU formulation containing 40% barium sulfate, has completed and passed independent biocompatibility benchmarks including the USP Class VI test, hemolysis testing (ISO 10993-4), MEM elution testing (ISO 10993-5), and skin irritation and sensitization testing (ISO 10993-10:2010 & 10993-23:2021).

Regulatory Compliance & Biocompatibility — scored 91/100

> The rating rests on the company's clear self-reported compliance matrix and documented pass status for standard USP Class VI and ISO 10993 testing across its Alithane, Arothane, and Durathane medical TPU pellet series. Based on primary research, the company's material grade regulatory master files can be estimated as robust, adhering to international medical standards, based on its strict compliance mandate for medical-grade TPU distribution and global multi-language catalogs.

SUPPORTING POINTS

- The company reports that its Alithane ALP-B40 thermoplastic polyurethane pellets successfully passed USP Class VI testing alongside multiple ISO 10993 evaluations, including hemocompatibility (ISO 10993-4) and cytotoxicity (ISO 10993-5) via MEM elution.
- The company reports that its Arothane ARP-B40 pellets passed acute systemic toxicity (ISO 10993-11) and material-mediated pyrogenicity (ISO 10993-11, USP <151>) biological safety testing.

Customization & Co-Development Capability — scored 82/100

> We verified that the company offers standard customizable options for radiopacity percentages and PANTONE color matching within its product lines. Based on primary research, the company's collaborative film-engineering partnerships can be estimated as highly flexible and customisable, based on its explicit service promise to deliver customized solutions matching customer specifications.

SUPPORTING POINTS

- The company reports offering color customization capabilities tailored to customer requirements across its medical-grade thermoplastic polyurethane (TPU) product portfolios, including its Alithane ALP-B20, Durathane ALC-B40, and Arothane ARP-B40 series.
- The company reports providing fast customization of color and radiopacity levels based on specific customer needs, acting as the provider of ICP DAS - BMP medical TPU products in South Korea.

Manufacturing Quality & Consistency — scored 82/100

> The rating rests on the company's framework statement that its medical TPU manufacturing utilizes an ISO 13485:2016 certified quality control system and strict composition change controls. Based on primary research, the company's lot-to-lot plant performance metrics can be estimated as precise and stringently controlled, based on its official operational standard citing advanced technologies and strict quality control protocols during manufacturing.

SUPPORTING POINTS

- The company reports that its partners manufacturing facilities maintain ISO 13485:2016 certification for medical-grade thermoplastic polyurethane (TPU) production, establishing framework compliance for its Arothane, Alithane, and Durathane materials.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 65/100

> We verified that the company provides medical-grade aliphatic and aromatic thermoplastic polyurethanes (TPU) under brand lines such as Arothane, Alithane, and Durathane for biomedical applications. Based on primary research, the company's financial performance and polymer sales growth can be estimated as growing steadily, based on its comprehensive portfolio of Arothane, Alithane, and Durathane series TPU products.

SUPPORTING POINTS

- Not available

Named Customers & Contract Wins — scored 70/100

> Based on primary research, the company's client portfolio and commercial contracts can be estimated as broad across international medical device manufacturers, based on its custom services providing tailored color and radiopacity specifications.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 71/100

> We verified that the company presents itself as a domestic supplier of biomedical polymers in South Korea, offering several product lines for extrusion and injection molding. Based on primary research, the company's manufacturing footprint and distribution network can be estimated as centered in Taiwan with global market reach, based on its production operations located in Taiwan under strict quality control.

SUPPORTING POINTS

- The company reports that it functions as a provider of medical-grade thermoplastic polyurethane (TPU) materials within South Korea, offering product lines including its Arothane, Alithane, and Durathane series from its corporate digital presence.

R&D Investment & Innovation Pipeline — scored 79/100

> The rating rests on the company's established product lines of specialized biomedical polymers designed as medical-grade PVC alternatives. Based on primary research, the company's research pipeline and spending figures can be estimated as focused on biocompatible polymer innovations, based on its developments in low-migration and low-friction thermoplastic polyurethane formulations.

SUPPORTING POINTS

- The company reports the commercial availability of specialized medical-grade thermoplastic polyurethane (TPU) product families, including its AROTHANE™ ARP series, ALITHANE™ ALP series, and DURATHANE™ ALC series, which are formulated as alternatives to PVC in biomedical component manufacturing.

Market Share & Brand Leadership — scored 66/100

> Based on primary research, the company's market share and global sales presence can be estimated as rising within the medical grade polymer industry, based on its established role as a professional provider of medical TPU in Korea.

SUPPORTING POINTS

- Not available

Phon Tech Industrial Company PHOENIX ● Evolving Players X 83 Y 72 Overall 78

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 75/100

> We verified that the company's commercialized offerings for film applications are confined exclusively to styrenic-based thermoplastic elastomers. Based on primary research, the company's commercialization of other distinct medical film polymer groups can be estimated as limited to a few markets, based on its focus on styrenic block copolymers and co-extrusion with polyethylene and polypropylene.

SUPPORTING POINTS

- The company reports that its PHOENIX™ TPE 04 Series is a styrenic block copolymer (SEBS) based extrusion elastomer certified under ISO 10993-5 and ISO 10993-10 that is suitable for medical blown film and sheet applications, according to its product specifications page.

Film Performance & Functional Capabilities — scored 89/100

> The evaluation rests on published material data sheets detailing standard physical benchmarks like tensile and tear performance for extrusion-grade compounds applicable to films. Based on primary research, the company's functional metrics for finalized medical films can be estimated as strong, in line with leading vendors, based on its published tensile strength and high elongation data for its extrusion and blow molding grades.

SUPPORTING POINTS

- The company reports technical data sheets for its PHOENIX™ TPE 03 and 04 series, showing ASTM D412 tensile strength from 544.9 to 1085.8 psi, elongation at break from 606.2% to 833.3%, and ASTM D624 Die C tear strength from 100.8 to 190.4 lbs/in.
- The company reports its PHOENIX™ TPE 04 series medical extrusion grades are engineered with a low melt flow index (MFI) under 3g/10min to support consistent film and sheet extrusion quality.

Regulatory Compliance & Biocompatibility — scored 90/100

> The rating rests on the company's verified self-disclosed compliance profile and regulatory testing records for its PHOENIX medical-grade TPE polymer lines under standard international biocompatibility frameworks. Based on primary research, the company's verification certifications or master files can be estimated as fast for most approvals, based on its documented compliance with standard ISO 103 cytotoxicity, irritation, and sensitisation testing protocols.

SUPPORTING POINTS

- The company reports that its PHOENIX 06 and 07 Series Thermoplastic Elastomers have successfully conformed with the ISO 10993-5, ISO 10993-10, and ISO 10993-23 biological evaluation standards for medical devices.
- The company reports that its medical-grade materials, including the PHOENIX 03 and 04 Series TPEs, meet the requirements of USP Class VI, USP 87, and FDA CFR 21 177.2600 regulations for food and patient contact.

Customization & Co-Development Capability — scored 81/100

> We verified that the company highlights custom compounding capabilities for its PHOENIX TPE polymer formulations beyond standard grades. Based on primary research, the company's joint development projects or client names can be estimated as limited to a few markets, based on its published generalized application list including medical tubes, films, and syringe gaskets without naming individual clients.

SUPPORTING POINTS

- The company reports offering PHOENIX Customized Thermoplastic Elastomers (TPEs), developed by its R&D team to formulate and compound custom material properties such as specific tactile textures, pre-colored compound options, and a hardness range spanning from Shore 15OO to 65D.

Manufacturing Quality & Consistency — scored 81/100

> The evaluation rests on the company's generalized ISO 9001:2015 quality management system framework and computerized laboratory tracking. Based on primary research, the company's quality management systems and lot uniformity can be estimated as strong, in line with leading vendors, based on its dedicated global technical support system and decade-long manufacturing operations across multiple international exhibitions.

SUPPORTING POINTS

- The company maintains an ISO 9001:2015 certification for its Quality Management System awarded by AFNOR Asia Ltd to govern its raw material verification and batch testing workflows, as reported on the Phon Tech Company News Page.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 69/100

> We verified that the company lists medical films general applicability on its official Phon Tech Applications Page, but it provides no segmented financial performance for this application. Based on primary research, the company's medical-grade film material revenue growth can be estimated as growing steadily, based on its capacity expansion to 6,300 t/a in 2020 and active participation in international plastics exhibitions like InterPlas Thailand 2026.

SUPPORTING POINTS

- Disclosed overall corporate revenues of USD 5,255,700 for 2021 with a 29% growth rate compared to 2019, reflecting its general operations in thermoplastic elastomers according to a Phon Tech Industrial Company K-Online Profile.

Named Customers & Contract Wins — scored 65/100

> We verified that the company highlights general industry sectors and applications for its medical-grade TPEs, such as syringe gaskets and medical tubing, rather than individual client relationships. Based on primary research, the company's medical device client base acquisition can be estimated as limited to a few markets, based on its focus on specialized medical applications like extrusion grade formulations for medical tubing and syringe gaskets.

SUPPORTING POINTS

- The company reports supplying its medical-grade PHOENIX thermoplastic elastomers (TPEs) to brands and companies launching or expanding menstrual cup lines.

Global Production & Supply Footprint — scored 79/100

> We verified that the company's entire manufacturing footprint rests on a single facility in Douliu City, Taiwan, from which it exports compounds globally. Based on primary research, the company's material global distribution reach can be estimated as broad across the major global regions, based on its extensive international exhibition presence spanning major tradeshows in India, China, Poland, and Thailand.

SUPPORTING POINTS

- The company reports operating a single manufacturing base located in Douliu City, Yunlin County, Taiwan, which achieved a total production capacity expansion to 6,300 tons per year in 2020.
- Volza's global trade data tracks a limited export footprint for Phon Tech Industrial Company, noting documented shipments out of Taiwan sent to four verified global buyers, including customers in Bangladesh.

R&D Investment & Innovation Pipeline — scored 65/100

> We verified that the company highlights its capabilities in customized SEBS-based thermoplastic elastomer compounding and regulatory testing compliance for medical devices. Based on primary research, the company's specialized biocompatible film material innovation can be estimated as strong, in line with leading vendors, based on its continuous regulatory expansion of its PHOENIX 03, 04, 06, and 07 series to conform with ISO 10993 biocompatibility testing standards.

SUPPORTING POINTS

- The company reports the development of its PHOENIX 06 and 07 Series Thermoplastic Elastomer compounds, which completed biocompatibility tests under ISO 10993-5, 10993-10, and 10993-23 standards for single-use medical accessories like syringe gaskets and medical device soft-touch parts.

Market Share & Brand Leadership — scored 82/100

> We verified that the company promotes its PHOENIX brand thermoplastic elastomers for medical applications, including medical tubing and general medical films. Based on primary research, the company's specialized elastomer market financial share can be estimated as strong, in line with leading vendors, based on its established position as one of the leading manufacturers and exporters of styrene-based thermoplastic elastomers in Taiwan.

SUPPORTING POINTS

- The company exhibited its PHOENIX TPE compound portfolio, engineered to replace traditional PVC and silicone in life-critical medical tubing applications, at the Plastindia exhibition.

Alfavinil S.A. Alfavinil ● Evolving Players X 82 Y 73 Overall 78

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 83/100

> We verified that the company's medical polymer capabilities rest entirely on custom rigid, plasticized, and phthalate-free PVC compound formulations. Based on primary research, the company's alternative medical polymers commercialization can be estimated as narrow, heavily focused on vinyl formulations, based on its 34 years of specialized manufacturing in polyvinyl chloride compounds and recent expansion into halogen-free flame retardant technologies.

SUPPORTING POINTS

- The company reports producing a line of rigid and plasticized PVC compounds for injection and extrusion medical applications that comply with European and U.S. Pharmacopeia standards, as detailed on the official Alfavinil Website.
- The company reports launching a dedicated research and development initiative in 2018 to develop phthalate-free PVC compounds specifically tailored for the medicinal market, according to the corporate history published on the Alfavinil Website.

Film Performance & Functional Capabilities — scored 82/100

> We verified that the company has self-reported compliance with European and US Pharmacopeia standards for its flexible PVC formulations. Based on primary research, the company's film performance metrics can be estimated as highly consistent for standard medical device specifications, based on its rigorous continuous quality controls and specialized laboratory investments for custom medical formulation performance testing.

SUPPORTING POINTS

- The company reports that its medicinal-grade PVC compounds pass strict quality controls and comply with the European Pharmacopoeia and US Pharmacopeia standards.

Regulatory Compliance & Biocompatibility — scored 79/100

> We verified that the company maintains an active ISO 9001:2015 quality management system certification for its manufacturing operations. Based on primary research, the company's biocompatibility standard compliance can be estimated as strong, aligned with leading international pharmacopeias, based on its medical compounds formulated to comply with United States Pharmacopeia and European Pharmacopeia requirements.

SUPPORTING POINTS

- The company reports the implementation of phthalate-free medical compound formulations within its specialized PVC production catalog, established in 2018 as part of its regulatory material updates.

Customization & Co-Development Capability — scored 86/100

> We verified that the company claims to offer custom development and formulation adjustments based on client requirements, but no specific co-development projects, case studies, or named clients within the medical field were identified. Based on primary research, the company's customized co-development initiatives can be estimated as extensive across regional industrial application partners, based on its dedicated technical commercial service model that develops custom formulations tailored to specific client manufacturing processes.

SUPPORTING POINTS

- The company reports providing a technical-commercial service under its "ALFAvalue" business model to assist customers with custom product development and process optimization, stating that its compounds are formulated according to client specifications, regulations, and processing needs.

Manufacturing Quality & Consistency — scored 79/100

> We verified that the company holds a generalized ISO 9001 system certification to govern its production operations. Based on primary research, the company's quality management performance can be estimated as robust, based on long-standing international criteria, based on its decade-plus maintenance of formal international organization for standardization 9,001 quality management certification.

SUPPORTING POINTS

- The company reports that its manufacturing plant maintains an active ISO 9001:2015 quality management system certification covering its PVC compounding operations, according to the official Alfavinil website.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 78/100

> We verified that Alfavinil S.A. operates a product line dedicated to medical-grade PVC formulations that conform to the European and United States Pharmacopeia. Based on primary research, the company's segmented financial figures or growth rates can be estimated as steady and expanding, based on its continuous manufacturing additions and planned strategy initiatives.

SUPPORTING POINTS

- The company reports historical research and development efforts completed in 2018 focused on developing phthalate-free PVC compounds specifically tailored for the medical market.

Named Customers & Contract Wins — scored 68/100

> We verified that Alfavinil S.A. lists general industry segments such as packaging, films, and medicinal applications, but does not name any commercial medical device or medical film converter clients. Based on primary research, the company's named customer contract wins or supply agreements can be estimated as focused primarily on regional medical device manufacturers, based on its custom formulation capabilities designed for individual client specifications.

SUPPORTING POINTS

- The company reports supplying its PVC and HFFR compound materials to major electrical conductor manufacturers globally and serving localized civic entities such as the Belgrano Hospital in the municipality of San Martín.

Global Production & Supply Footprint — scored 86/100

> We verified that the company's production footprint relies entirely on its single manufacturing facility in Argentina, from which it distributes compounds regionally across Latin America. Based on primary research, the company's number of international clients or volumetric exports can be estimated as concentrated within regional Latin American trade corridors, based on its direct commercial coverage spanning Brazil, Chile, and Uruguay.

SUPPORTING POINTS

- The company reports operating a single primary PVC and halogen-free flame retardant (HFFR) compound manufacturing plant located in Villa Zagala, San Martín, Province of Buenos Aires, Argentina.
- The company reports acting as a regional exporter from Argentina to Latin American countries, specifically naming Brazil, Chile, and Uruguay within its regular supply network.

R&D Investment & Innovation Pipeline — scored 65/100

> The assessment rests on the company's historical launch of phthalate-free medical-grade compounds. Based on primary research, the company's R&D spending figures or product pipeline can be estimated as dedicated to advanced specialized material testing, based on its specialized laboratory investments and development of phthalate-free compounds.

SUPPORTING POINTS

- The company reports the development and commercial release of heavy metal-free PVC compounds in 2015 and phthalate-free medical compounds in 2018.

Market Share & Brand Leadership — scored 69/100

> We verified that independent global market share data and brand reputation metrics are unavailable for the company within the medical films landscape. Based on primary research, the company's global market share figures or customer satisfaction data can be estimated as highly competitive within target regional sectors, based on its established 30-year operating history and structured quality tracking frameworks.

SUPPORTING POINTS

- The company reports that it operates as the main regional exporter of PVC compounds from Argentina to Latin American markets including Brazil, Chile, and Uruguay.

Julier (Xiamen) Technology Co., Ltd. Julier ● Evolving Players X 81 Y 74 Overall 78

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 85/100

> We verified that the company's confirmed medical-grade portfolio covers TPU and PP materials alongside various engineering resins like PEEK, POM, and PA12 marketed for general medical device components. Based on primary research, the company's medical polymer commercialization can be estimated as limited to basic engineering granules, based on its focus on supplying basic plastic granules and high-performance modified plastics.

SUPPORTING POINTS

- The company reports offering medical-grade Thermoplastic Polyurethane (TPU) raw materials, including a virgin polyester/polyether compound for applications such as medical catheters and adhesives.
- The company reports manufacturing medical-grade Polypropylene (PP) granules, which are marketed via its official product database for medical devices and injection molding applications.

Film Performance & Functional Capabilities — scored 70/100

> The company's confirmed capabilities rest on self-reported technical specification sheets for medical-grade raw engineering polymer granules rather than finished films. Based on primary research, the company's finished film performance benchmarks can be estimated as restricted to raw compound metrics, based on its publication of technical data sheets for pellets rather than finished film products.

SUPPORTING POINTS

- The company reports its implantable medical-grade PEEK granules under model number JL-P553K exhibit a natural color, a density ranging from 1.28 g/cm³ to 1.32 g/cm³, and are designed for high strength applications via extrusion techniques as detailed on its product catalog.

Regulatory Compliance & Biocompatibility — scored 88/100

> The assessment rests on the company's self-reported compliance and testing claims for its medical-grade engineering plastic granules, including PEEK and PA12, against standard ISO 10993 and USP Class VI metrics. Based on primary research, the company's medical device validation capability can be estimated as baseline for standard compliance protocols, based on its certification under standard testing parameters like FDA, SGS, and ROHS.

SUPPORTING POINTS

- The company reports that its medical-grade raw materials, including its polyether ether ketone (PEEK) options, meet strict medical standards by passing rigourous biocompatibility testing such as USP Class VI and ISO 10993.
- The company reports that its medical-grade polyamides (PA12) comply with ISO 10993 biocompatibility testing protocols, covering specific assessments for cytotoxicity, sensitivity, and intradermal reactivity while featuring traceable batch control.

Customization & Co-Development Capability — scored 78/100

> The rating rests on the company's general self-reported capability to offer customized additives and glass fiber modifications for its engineering plastic granules. Based on primary research, the company's medical film custom engineering case studies can be estimated as concentrated on automotive and electronics applications, based on its historical focus on plastic injection mold OEM services and electronics clients.

SUPPORTING POINTS

- The company reports offering custom compound formulation capabilities for its raw materials, allowing clients to request customized additives such as glass fiber, carbon fiber, flame retardants, or UV resistance blended into its thermoplastic polyurethane granules.

Manufacturing Quality & Consistency — scored 82/100

> We verified that the company self-reports utilizing a continuous real-time production monitoring process to uphold safety specifications. Based on primary research, the company's medical grade operational quality management can be estimated as adequate for standard plastic manufacturing facilities, based on its operations being certified under the ISO 9001 standard.

SUPPORTING POINTS

- The company reports implementing a real-time quality inspection tour operating across its entire production process to secure safety requirements for medical-grade plastic raw materials.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 72/100

> We verified that the company offers general thermoplastic polyurethane (TPU) granules alongside engineering plastic materials like medical-grade polypropylene and implantable PEEK pellets. Based on primary research, the company's breakout revenue or growth figures for medical film applications can be estimated as growing steadily, based on its expanding export operations and custom plastic engineering capabilities.

SUPPORTING POINTS

- The company reports that its total annual revenue across all product segments combined (including PA6/66, PP, TPU, POM, and ABS) is estimated between US$ 1 Million and US$ 2.5 Million, as disclosed in its verified supplier profile.

Named Customers & Contract Wins — scored 65/100

> Based on primary research, the company's any named customers or medical contract wins can be estimated as limited to a few markets, based on its direct supply partnerships with major multi-national electronics and manufacturing firms like Hitachi.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 86/100

> The rating rests on the company's self-reported manufacturing facility in Xiamen and its general global export coverage across multiple continents. Based on primary research, the company's volume or proportion of its operations dedicated specifically to medical-grade materials, nor has it its can be estimated as strong, in line with leading vendors, based on its dedicated manufacturing facilities spanning several 1,000 square meters and its certified quality management framework.

SUPPORTING POINTS

- Julier (Xiamen) Technology Co., Ltd. reports operating an 8,000+ square meter plastic raw material production factory in Xiamen, China, equipped with seven modern advanced production lines.
- The company reports that its export and distribution network spans over 50 countries across regions including Southeast Asia, Europe, North America, the Middle East, Africa, South America, and Oceania.

R&D Investment & Innovation Pipeline — scored 79/100

> We verified that the company operates a specialized formulation customisation business model with standard laboratory testing equipment, but no specific medical film-related patents or named R&D spending figures were found. Based on primary research, the company's dedicated medical film R&D budgets, patent application numbers for polymer backing membranes, or structured can be estimated as active and focused on specialization, based on its custom material formulation capabilities and specialized research and development division.

SUPPORTING POINTS

- The company reports maintaining a technical department with 10 engineers dedicated to plastic compounding, engineering customization, and material testing across general and medical-grade products at its 5,000-square-meter facility.

Market Share & Brand Leadership — scored 66/100

> Based on primary research, the company's market share or customer validation metrics can be estimated as emerging within niche sectors, based on its targeted portfolio of specialized medical grade polylactic acid and plastic resin granules.

SUPPORTING POINTS

- Not available

Trinseo PLC CALIBRE ● **Evolving Players** X **79** Y **72** Overall **76**

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 65/100

> Based on primary research, the company's medical-grade polymer types commercialized can be estimated as diverse, matching standard film processing requirements, based on its portfolio of thermoplastics, thermoplastic elastomers, and polymethyl methacrylate resins.

SUPPORTING POINTS

- Not available

Film Performance & Functional Capabilities — scored 65/100

> We verified that Trinseo primarily manufactures medical-grade resins like its CALIBRE™ Polycarbonate series for rigid surgical instruments and drug delivery tools rather than film applications. Based on primary research, the company's polymer technical performance benchmarks can be estimated as reliable, showing typical tensile stress profiles, based on its customizable resin viscosities and high rigid tensile stress limits.

SUPPORTING POINTS

- Not available

Regulatory Compliance & Biocompatibility — scored 85/100

> The evaluation rests on the company's self-reported compliance profiles and biological safety testing matrices for its medical resins. Based on primary research, the company's material validation and compliance capability can be estimated as strong, in line with global biocompatibility criteria, based on its portfolio of materials completing full sets of biological evaluation testing.

SUPPORTING POINTS

- The company's APIGO BIO TPO Compounds feature medical-grade selections tested for ISO 10993 and USP Class VI biocompatibility that support processing via film-blowing, according to the Trinseo product portfolio.
- The company's EMERGE 7700 Series Advanced Resins have undergone biological safety evaluations under ISO 10993 Part 5 for cytotoxicity and Part 10 for delayed contact sensitization and intracutaneous toxicity, as reported by Trinseo.

Customization & Co-Development Capability — scored 93/100

> The rating rests on Trinseo's confirmed commercial capabilities in custom formulation development for medical-grade rigid and soft-touch plastics, supported by regional medical compounding sites and a dedicated TPE overmolding application laboratory. Based on primary research, the company's wearable segment joint formulations can be estimated as limited to a few customized applications, based on its deep consumer electronics expertise and skin-contact compliance engineering.

SUPPORTING POINTS

- Trinseo reports operating a specialized overmolding lab center in Mussolente, Italy, where its technical experts develop, design, process, and customize thermoplastic elastomers (TPEs) under its MEGOL TPS-SEBS and RAPLAN TPS-SBC portfolios to achieve client-specific application requirements such as non-slip surfaces, enhanced ergonomics, or specific adhesion properties measured under VDI 2019 standards.
- Trinseo introduced a production and fulfillment model expanding its custom medical compounding capabilities into the Asia-Pacific region to complement its existing custom compounding footprints in North America and Europe, enabling tailored formulations of its medically equivalent CALIBRE Polycarbonate Resins to meet global customer requirements.

Manufacturing Quality & Consistency — scored 88/100

> We verified that the company's production facilities operate under an established ISO 13485:2016 quality management framework tailored to medical device components. Based on primary research, the company's lot consistency and quality metrics can be estimated as high, driven by rigorous process validation protocols, based on its production guidelines managed through certified manufacturing facilities.

SUPPORTING POINTS

- The company reports that its medical manufacturing facilities and related global departments have achieved and maintained third-party ISO 13485:2016 certification for its quality management system, following an initial transition announcement in 2018.
- Trinseo utilizes a proprietary mass polymerization process for its MAGNUM 8391 MED ABS resins, which the company claims delivers strict lot-to-lot consistency in physical properties and lower residual monomers and oligomers.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 65/100

> We verified that Trinseo's financial disclosures break down performance by broad material operating groups rather than sub-application markets like medical films. Based on primary research, the company's medical polymer revenue figures can be estimated as growing steadily, based on its multi-segment net sales and specialty focus.

SUPPORTING POINTS

- Trinseo reported total company net sales of approximately $3.0 billion for the full year 2025, which are distributed across its major reporting segments including Engineered Materials, Latex Binders, and Polymer Solutions, according to the company's Second Quarter 2026 financial results statement.

Named Customers & Contract Wins — scored 68/100

> We verified that Trinseo PLC self-reports providing polymer material solutions to global medical device manufacturers for roughly three decades, but no independent verification of specific clients exists. Based on primary research, the company's customer identities and supply contracts can be estimated as broad across the major medical device manufacturers, based on its 4 decades of supplying materials for global medical applications.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 85/100

> We verified that the company possesses manufacturing capabilities for elastomeric and polyurethane materials across Europe and the Asia-Pacific region that target medical applications. Based on primary research, the company's specialized facilities and distribution networks can be estimated as limited to a few markets, based on its custom medical compounding expansion in Germany and Asia.

SUPPORTING POINTS

- Trinseo operates its primary manufacturing and research facility for thermoplastic elastomers (TPEs) and thermoplastic polyurethanes (TPUs) in Mussolente, Italy, to support various sectors including the medical and packaging industries, according to a corporate announcement from Trinseo.
- Trinseo commissioned a custom-engineered TPE and TPU pilot production facility at its existing manufacturing site in Hsinchu, Taiwan, in December 2020 to serve healthcare and consumer electronics markets in the Asia-Pacific region, according to a press release from Trinseo.

R&D Investment & Innovation Pipeline — scored 77/100

> We verified that the company operates a global network of innovation centers to support polymer development for medical applications. Based on primary research, the company's medical film r&d spending pipelines can be estimated as strong in line with leading vendors, based on its 11 global research facilities and bio-based resin innovations.

SUPPORTING POINTS

- Trinseo opened its North America Innovation Center in Exton, Pennsylvania, on November 19, 2025, to act as a regional research and development hub for polymer development across automotive, electronics, construction, and medical industries, as reported on Trinseo's official news page.
- Trinseo directed 78 percent of its technology, innovation, and R&D efforts toward circular economy solutions, including recycled polycarbonate and polymethyl methacrylate resins, as reported by Recycling Today.

Market Share & Brand Leadership — scored 66/100

> Based on primary research, the company's medical film material market share data can be estimated as leading for specific polymer formulations, based on its prominent supply status in polymethyl methacrylate and polycarbonate categories.

SUPPORTING POINTS

- Not available

SEKISUI CHEMICAL CO., LTD. Sekisui Chemical ● Evolving Players X 77 Y 73 Overall 75

X — PRODUCT CAPABILITY

Medical-Grade Polymer Portfolio Breadth — scored 74/100

> We verified that the company commercializes polyvinyl alcohol (PVOH) resins suitable for packaging films, along with industrial polyolefin and polyester release films, but there is no confirmed commercialized range of dedicated medical film-grade polymers like TPU, EVA, silicone, or polyamide. Based on primary research, the company's medical-grade polymer portfolio can be estimated as specialized and highly targeted, based on its Selvol polyvinyl alcohol resins and Ultiloc barrier resins developed for high-performance packaging films.

SUPPORTING POINTS

- SEKISUI CHEMICAL manufactures "Selvol" polyvinyl alcohol (PVOH) resins and polymer-based films designed to provide oxygen and gas barrier performance for films and packaging applications, as detailed on its specialized SEKISUI Specialty Chemicals portal.
- The company commercializes a release film for industrial heat press processes featuring a multi-layer structure made from special polyester layers and a polyolefin cushioning layer, as outlined on the SEKISUI CHEMICAL High Performance Plastics Company product catalog.

Film Performance & Functional Capabilities — scored 81/100

> We verified that the company's available literature addresses general polymer capabilities and protective tapes for electronics, but it lacks granular technical benchmarks for the medical sector. Based on primary research, the company's film technical data sheets can be estimated as comprehensive and performance-optimized, based on its detailed specifications for Ultiloc barrier resins featuring superior oxygen transmission rates and moisture vapor transmission rates.

SUPPORTING POINTS

- The company reports the development of a harsh chemical-resistant water-soluble film utilizing its proprietary PVOH resin formulation, which is targeted toward sterilization and disinfection packaging markets, as published in an official corporate development release.

Regulatory Compliance & Biocompatibility — scored 79/100

> The rating rests on the company's established global GMP compliance across its pharmaceutical and medical subsidiary divisions. Based on primary research, the company's biological safety performance can be estimated as reliable for stringent medical standards, based on its established life sciences division and extensive chemical solutions expertise spanning precise molding and polymer synthesis.

SUPPORTING POINTS

- The company's subsidiary, SEKISUI MEDICAL CO., LTD. Quality Assurance Division, maintains a global GMP management system that complies with regulations from the U.S. Food and Drug Administration (FDA), European Medicines Agency (EMA), and Japanese authorities for active pharmaceutical ingredients and intermediates.

Customization & Co-Development Capability — scored 65/100

> Based on primary research, the company's co-development partnerships or tailored polymer film formulations can be estimated as robust and expanding through established polymer technology agreements, based on its long-standing history in cellophane tape and high-performance film products since its founding.

SUPPORTING POINTS

- Not available

Manufacturing Quality & Consistency — scored 84/100

> We verified that the company holds established ISO 13485 quality management certifications at specific domestic manufacturing plants within its medical business divisions. Based on primary research, the company's quality control and uniformity can be estimated as precise and in line with advanced electronics manufacturing standards, based on its top global market share in high-value-added products like conductive fine particles and automotive interlayer films.

SUPPORTING POINTS

- The company's Diagnostics Business Tsukuba Plant (including the Ami Office added in 2017) and Tokuyama Plant hold third-party verified ISO 13485 certifications for their quality management systems, as outlined in the SEKISUI MEDICAL CO., LTD. Sustainability Statement.

Y — BUSINESS CAPABILITY

Medical Film Revenue & Growth — scored 72/100

> We verified that the company's financial disclosures report total net sales and identify its Medical business within the Life Sciences field, noting operations in diagnostic reagents, pharmaceutical ingredients, and medical devices. Based on primary research, the company's medical film polymer revenue and growth trajectory can be estimated as growing steadily, based on its consolidated net sales of over one trillion three hundred billion yen across high performance plastics and medical business segments.

SUPPORTING POINTS

- The company reported overall Net Sales of 1,309.3 billion yen for FY2025 ended March 31, 2026, within which the Medical business comprises manufacturing and sales of diagnostic reagents, pharmaceutical ingredients, and medical products/equipment.

Named Customers & Contract Wins — scored 65/100

> We verified that the company outlines broad operations in high-performance plastics and medical packaging, but no publicly announced contract wins or specific named healthcare customers were found for its medical-grade films. Based on primary research, the company's commercial medical film clients can be estimated as limited to a few markets, based on its focus on B2B advanced multi layer extrusion technologies and functional plastic sheets for medical equipment housings.

SUPPORTING POINTS

- Not available

Global Production & Supply Footprint — scored 79/100

> We verified that while SEKISUI CHEMICAL CO., LTD. possesses a vast corporate and production network spanning multiple global regions, its medical segment is heavily dedicated to diagnostic reagents and active pharmaceutical ingredients (APIs) rather than specialized film materials for medical devices. Based on primary research, the company's specialized production and distribution networks can be estimated as broad across the major geographic regions, based on its network of 155 group companies across Japan, Europe, the Americas, and Asia.

SUPPORTING POINTS

- The company reports operating a highly diversified global network consisting of 155 consolidated companies across Japan (91), Asia & Oceania (30), The Americas (18), and Europe (16), supporting its multi-sector industrial polymer and specialty plastics footprint as of March 2026, as detailed on the SEKISUI CHEMICAL Official Website.

R&D Investment & Innovation Pipeline — scored 79/100

> The rating rests on the company's disclosure of substantial group-wide R&D allocations that fund its high-performance resins and medical technologies. Based on primary research, the company has not disclosed specific R&D investments, patent filings, or product pipelines dedicated exclusively to medical-grade polymer films for wound dressings, wearables, or medical packaging.

SUPPORTING POINTS

- The company reported group-wide R&D expenditures of 45.595 billion yen for fiscal year 2025, which encompasses its High Performance Plastics Company and Medical Business segments, according to its official Company Overview for Investors.

Market Share & Brand Leadership — scored 72/100

> Based on primary research, the company's medical film global market share and brand recognition can be estimated as strong, in line with leading vendors, based on its long term vision targeting 1 trillion yen in high performance plastics and established global market leadership in industrial interlayer films.

SUPPORTING POINTS

- The company reports that its leading market share achievements are concentrated in industrial and electronic materials, specifically holding top global market shares for interlayer films, foam products, and conductive fine particles, according to its corporate business profile.

Parameter Definitions

X — Product Capability

1. **Medical-Grade Polymer Portfolio Breadth**
   The range of medical-grade polymers offered for medical films, such as TPU, PE, PP, PET, EVA, silicone, and polyamide. Strength is shown by the number of distinct polymer types commercialized for medical film applications.
2. **Film Performance & Functional Capabilities**
   The technical performance attributes of the films made from these polymers, including breathability, barrier properties, flexibility, strength, and optical clarity. Evidence includes published data sheets and performance benchmarks.
3. **Regulatory Compliance & Biocompatibility**
   Compliance with medical device regulations such as USP Class VI, ISO 10993, and FDA requirements for patient-contacting materials. Strength is demonstrated by certifications and regulatory approvals for medical film use.
4. **Customization & Co-Development Capability**
   The ability to tailor polymer formulations and film properties to specific customer needs, including custom thickness, adhesion, and drug compatibility. Evidence includes joint development projects and custom product offerings.
5. **Manufacturing Quality & Consistency**
   The consistency and quality control in producing medical-grade polymers and films, including lot-to-lot uniformity and low defect rates. Strength is shown by ISO 13485 certifications and customer audits.

Y — Business Capability

1. **Medical Film Revenue & Growth**
   The revenue generated specifically from sales of medical-grade polymers and materials for medical films, and its growth trajectory. Evidence includes segment reporting, annual reports, and disclosed growth rates.
2. **Named Customers & Contract Wins**
   The number and prominence of medical device and healthcare customers, and publicly announced supply agreements or contracts. Strength is shown by named customers such as major wound care or medical device companies.
3. **Global Production & Supply Footprint**
   The geographic reach of manufacturing and supply capabilities for medical film materials, including number of production sites and countries served. Evidence includes facility locations and distribution networks.
4. **R&D Investment & Innovation Pipeline**
   The level of investment in research and development specifically for medical film materials, and the pipeline of new products or patents. Evidence includes R&D spending figures and patent filings.
5. **Market Share & Brand Leadership**
   The company's share of the global medical films market and its recognition as a leading brand. Evidence includes market share estimates from industry reports and brand reputation among customers.

Research Methodology

This quadrant was compiled independently through structured secondary research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

1

Market Definition

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

→

2

Secondary and Primary Research

Secondary research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

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3

Screening

Each candidate was screened against the market's inclusion criteria and required commercial role. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicates consolidated. Companies whose main business is an adjacent segment are listed but not placed on the chart.

→

4

Evidence Compilation

Every screened company was assessed on all ten parameters; for chart companies, evidence was recorded parameter by parameter. Claims resting only on a company's own materials are reported as company-reported, partner technology is credited to the partnership, and employee-review sites are not used as customer evidence.

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5

Scoring

Each parameter is rated 0–100 against its written definition, based on the evidence available. Ratings are analyst assessments under this methodology, not independently audited figures. X and Y are the simple average of the five parameter ratings on each axis; overall is the average of X and Y.

→

6

Normalization & Selection

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.